NASA TM-100991 · BALLIN, 1988

Replicating a real-time turboshaft engine model

The T700-GE-700 engine and its fuel control, rebuilt from the report and checked against every comparison it prints

October 2026

What this presentation covers

1. **The source and the engine**
   Ballin's report, the T700, and what replication means
2. **One engine, three models**
   Simple nonlinear, real-time and linear, and how they relate
3. **How the model is built**
   Gas path and its thermodynamics, volumes, spools, heat soakage, the real-time frame, fuel control, sensors, data from a scan
4. **How the engine is run**
   Steady state, open loop, closed loop, manoeuvres, linearization
5. **How closely it matches**
   Every comparison the report prints
6. **Findings, decisions and limits**
   What the report never printed, where it is wrong, and where we differ

The source: NASA TM-100991 (Ballin, 1988)

- **Ballin, M. G.**, *A High Fidelity Real-Time Simulation of a Small Turboshaft Engine*, NASA TM-100991, NASA Ames Research Center, July 1988.
- **Purpose:** an engine model fast enough for a piloted helicopter simulator. It flew with the Gen Hel UH-60A blade-element rotor simulation on a CDC 7600 [pdf p.36].
- **What it gives us:**
  - Eqs. 1–80: the engine model and its real-time solution
  - Appendix A: 33 constants and 11 engine functions (the maps), printed as plots
  - Appendix B: 12 linear models (A and b matrices) at three flight conditions
  - Appendix C: the complete fuel control: 22 block diagrams and 8 schedules, no equations
  - Validation: Table B.1 (trim states), Table 1 (eigenvalues), Figures 6–10 (steady sweeps, fuel transients), Figures 4–5 (linear responses), Table 3 (a prototype test engine), Figures 11–15 (the closed loop inside the helicopter)

Why the engine matters to a flight simulator

- The pilot feels the engine through **rotor speed**. Fuel control, drive train and rotor interact at low frequency (Figure 1, right).
- A simulator must keep that band right, **in real time**: Ballin's frame limit was 10 ms [pdf p.38].
- Faster physics, such as gas filling and emptying the engine's internal volumes, can be approximated as long as that band survives.

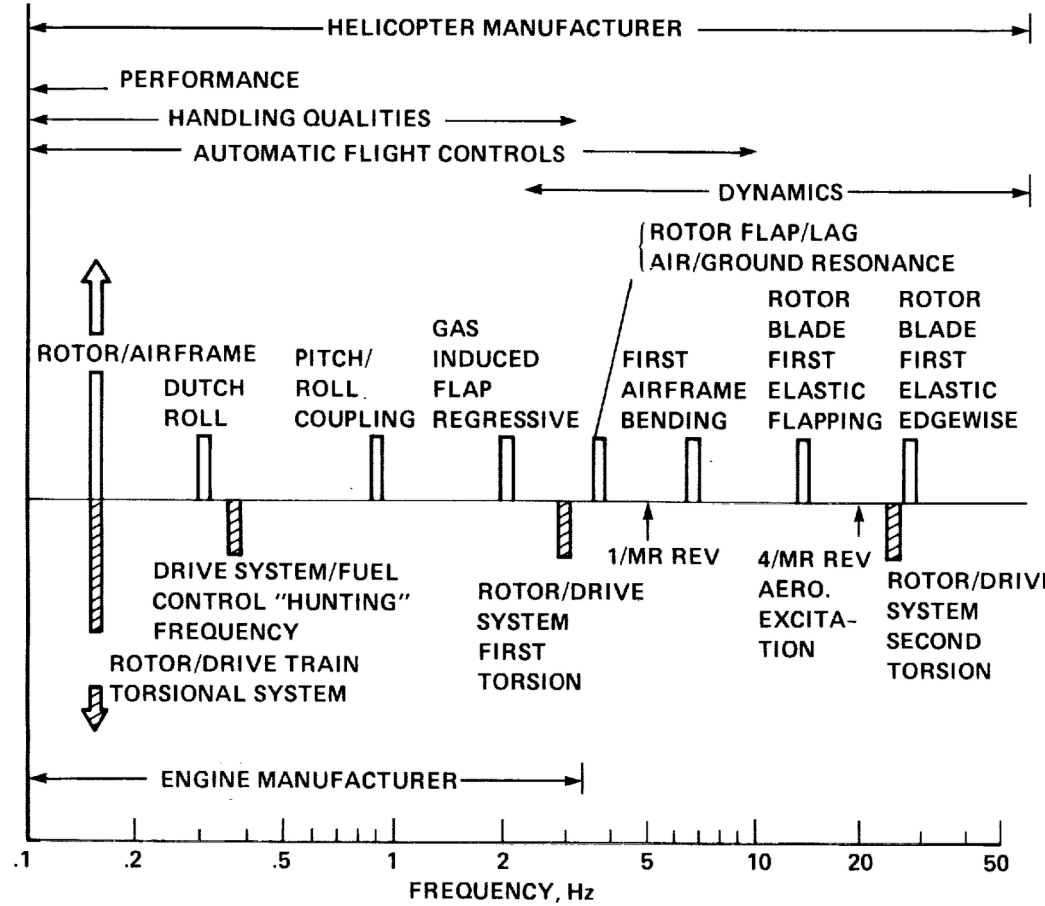


Figure 1: Modal frequencies of interest in engine and fuel-control design and modeling (ref. 3).

Ballin, Figure 1 (from his ref. 3), pdf p.16

The engine: General Electric T700-GE-700

- A small turboshaft of the **1600-hp class** that powers the UH-60A Black Hawk; the model is one engine.
- **Compressor:** five axial stages and one centrifugal stage, with variable inlet guide vanes and stator vanes on the first two stages.
- **Combustion chamber:** annular and flow-through, at low fuel pressure.
- **High-pressure turbine** (the report's gas-generator turbine): two axial stages, cooled by air bled from the compressor exit; drives the compressor at speed **NG**.
- **Power turbine:** two stages, independent and uncooled; its coaxial shaft runs forward through the engine to the output, at speed **NP**.

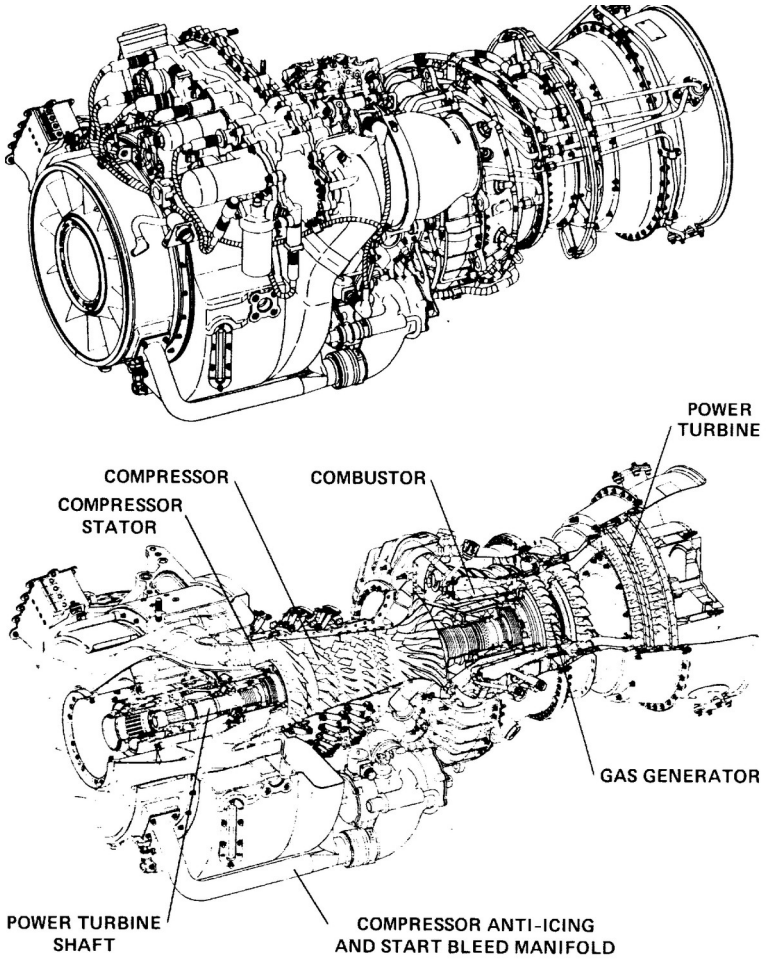


Figure 2: General Electric T700-GE-700 engine. (From ref. 6.)

Ballin, Figure 2 (from his ref. 6), pdf p.18; description pdf p.17

Stations and components

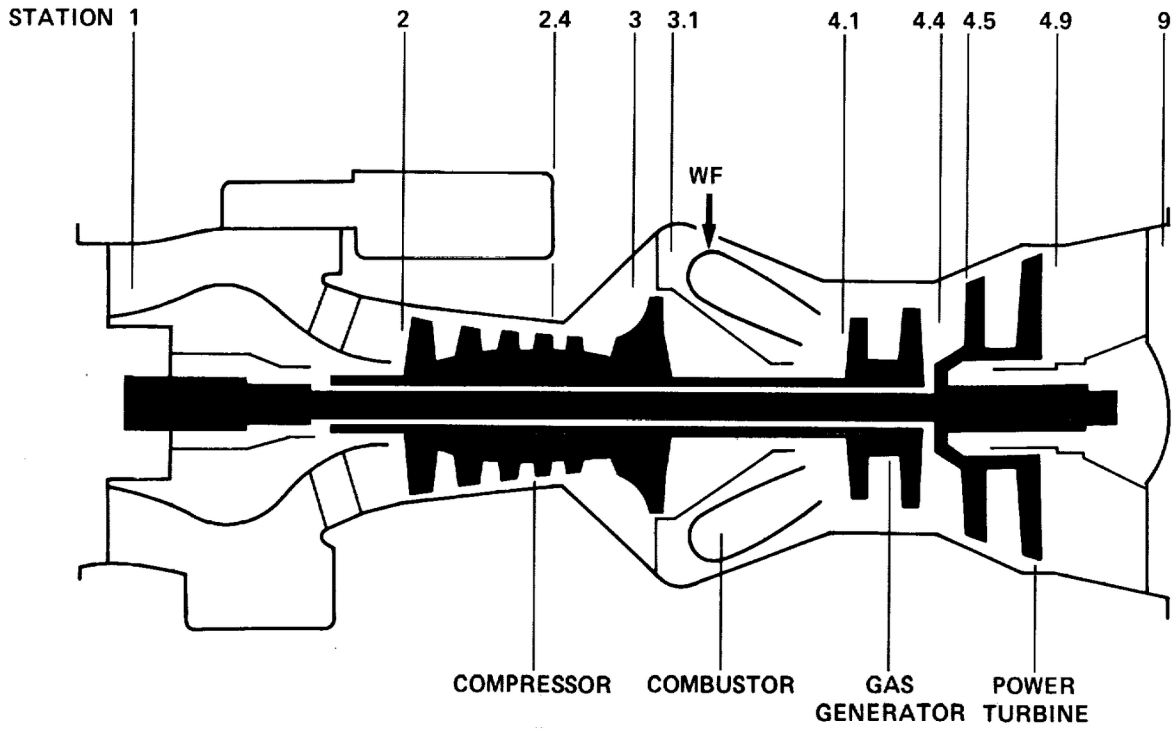


Figure 3: Primary components of a small turboshaft engine.

Ballin's Figure 3: primary components of a small turboshaft engine. WF marks the fuel entering the combustor.

Station	Where
2	compressor face
2.4	stage-4 bleed: seal pressurization, power-turbine balance
3	compressor diffuser exit; bleed to cool combustor and turbine
3.1	combustor
4.1	combustor exit = gas-generator turbine inlet (a mixing volume)
4.4	gas-generator turbine exit, before the cooling air returns
4.5	power-turbine inlet, cooling air mixed back in
4.9	power-turbine exit
9	exhaust

There is no station 4 or 5.

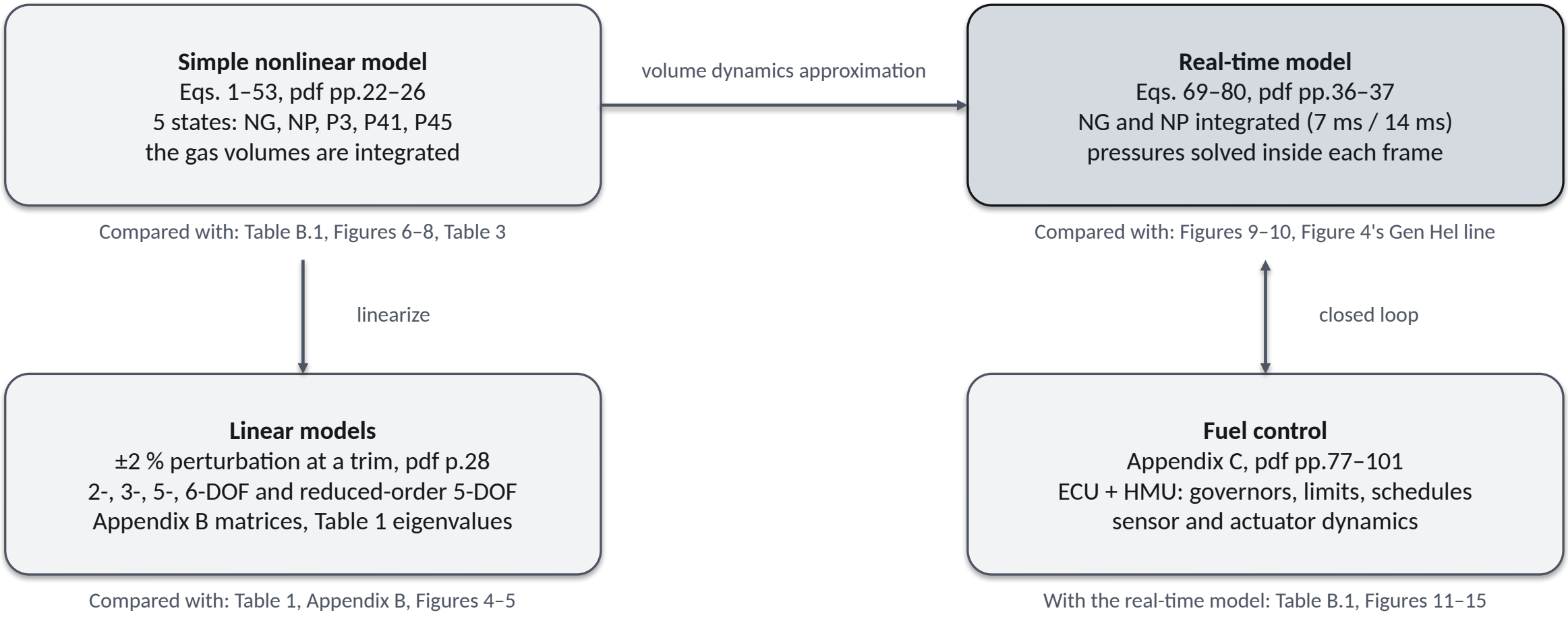
Ballin, Figure 3 and text, pdf pp.20–21

What “replication” means here

- **Reproduce the report as written**, including its simplifications and its admitted biases. Not an improvement.
- **Every number is traced to a page** of the report and read from the page image; the scan's text layer corrupts digits and symbols.
- **Where the report is silent**, we decide, record why, and measure what the choice changes.
- **Where following it literally fails**, we depart from it and measure the cost. There are four such departures (Part 6).
- **Never tuned to pass:** no constant is adjusted to improve agreement. Acceptance bounds come from the scan's measured read error.
- **An example of what not to fix:** Ballin's model reads turbine temperature slightly low at trim in every validation case, and he says so [pdf p.47]. Reproducing that bias is correct.
- Built in Python with NumPy, in the report's own units (lbm, psia, °R, hp, rpm) and symbols.

Backup slide, hidden in the show

One engine, three models



Steady states are the same in the simple nonlinear and the real-time model: the approximation changes only how the engine moves between them.

Why the real-time model drops the volume dynamics

Mode 5-DOF, hover	Eigenvalue 1/s	Time constant	Corner frequency |λ|/2π, Hz
NP	−0.565	1.77 s	0.09
NG	−2.66	0.38 s	0.42
P3	−51.6	19.4 ms	8.2
P45	−3,060	0.33 ms	487
P41	−4,900	0.20 ms	780

- **Euler integration**, the simplest explicit step, is stable only for $\Delta t < 2/|\lambda|$: about **0.4 ms** for the fastest mode. Ballin's engine frame is **7 ms**.
- The spools are about a thousand times slower than P41 and P45. So: integrate only NG and NP, and solve P3, P41, P45 inside each frame as if the volumes filled instantly.
- Conclusions: “The modeling of high-speed dynamics which represent changes of mass flow between the internal control volumes was found to be unnecessary” [pdf p.54].

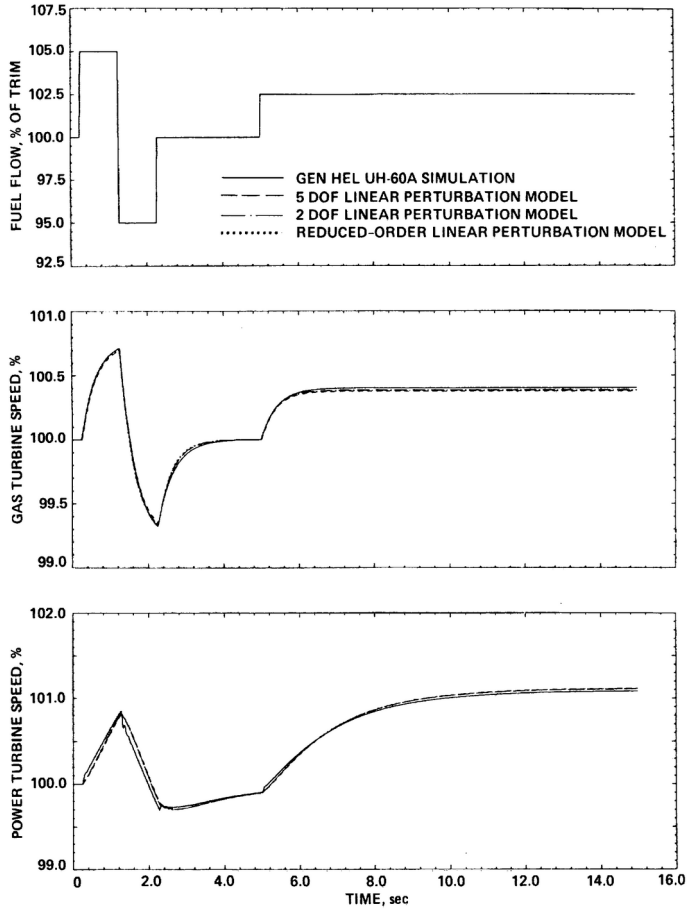


Figure 4: Comparison of the linear perturbation models with the nonlinear simulation for a high-power flight condition.

Ballin's Figure 4: linear models against the nonlinear simulation in Gen Hel. The 2-DOF model has no volume dynamics, and tracks the others.

Table 1, pdf p.31; Ballin, Figure 4, pdf p.30; frame, pdf p.47; Conclusions, pdf p.54

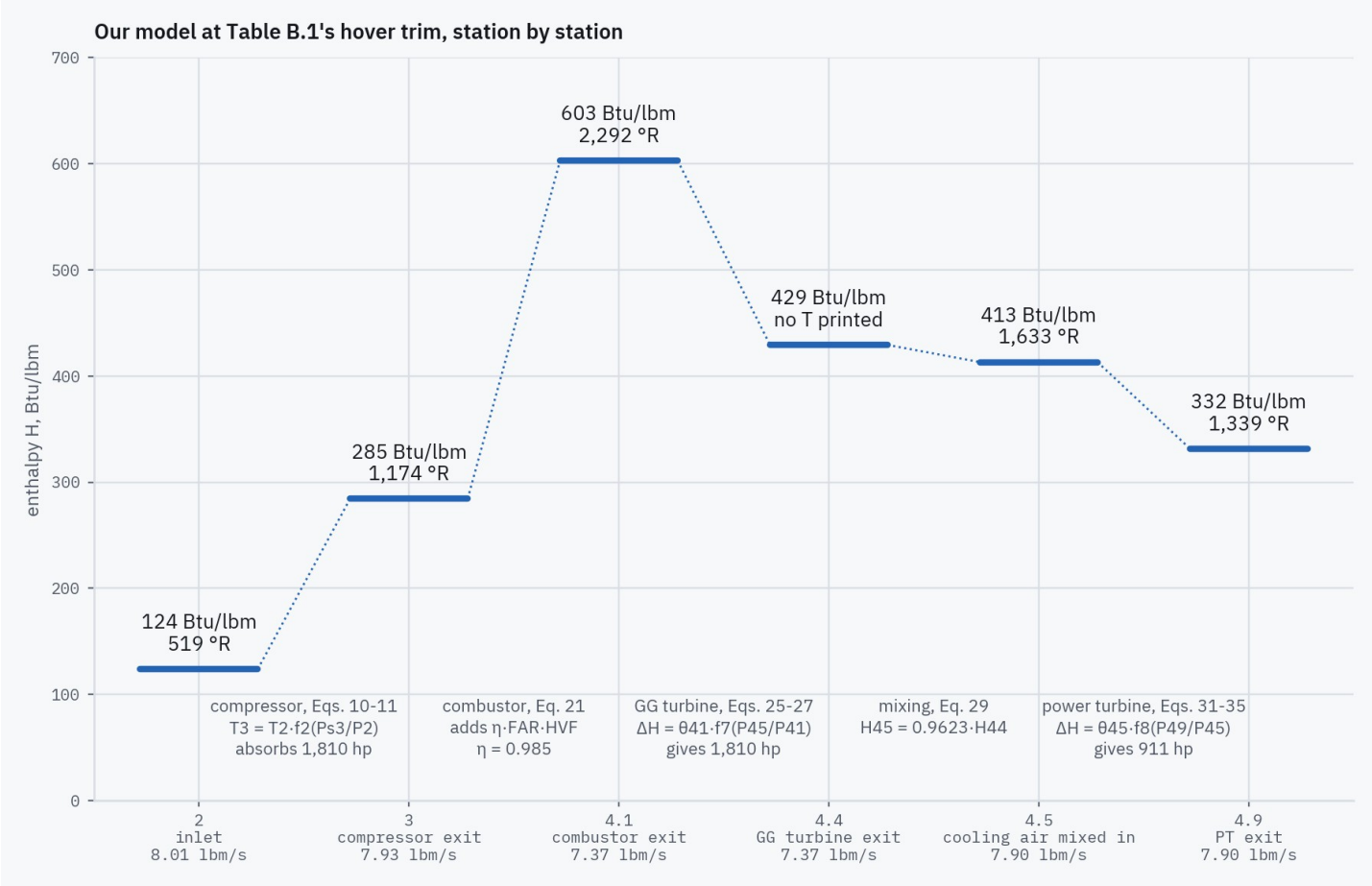
The gas path, component by component

Component	What the model computes	Report function (plot)
Inlet	station 2 pressure and temperature from ambient	—
Compressor	corrected airflow from the map, against Ps3/P2 and corrected speed; exit temperature	f1 (Fig. A1), f2 (A2)
Bleeds	seal pressurization and power-turbine balance (stage 4); impeller-tip leakage and turbine cooling (diffuser)	f3, f4, f5 (A3–A5)
Combustor	efficiency, exit temperature T41, pressure drop	f6 (A6)
Gas-generator turbine	enthalpy drop against P45/P41; choked flow W41	f7 (A7)
Station 4.5	turbine cooling air mixed back in	—
Power turbine	enthalpy drop against P49/P45; flow against Ps9/P45	f8, f9 (A8, A9)
Exhaust	pressure loss against corrected speed	f10 (A10)
Torques	compressor, gas-generator turbine, power turbine (with a speed-damping term)	—

Gas properties come from the report's own straight-line enthalpy–temperature fits (Table A.1; next slide).

Eqs. 1–41, pdf pp.22–25; functions f1–f10 from Appendix A, pdf pp.56–65

How the energy is carried: Ballin's thermodynamics



- **No gas tables.** At each station, enthalpy is a straight line in temperature: a fixed specific heat, 0.239 to 0.301 Btu/lbm·°R.
- **Compressor:** exit temperature from pressure ratio alone (f2).
- **Turbines:** enthalpy drop = θ × f(pressure ratio), θ a straight line in inlet temperature (f7, f8).
- No efficiency maps: the efficiencies are built into f2, f7 and f8.
- **Combustor:** fuel energy = 18,300 Btu/lbm × efficiency 0.985.
- **Cooling air** returns as a fixed fraction: H45 = 0.9623·H44. At hover 1.5 % of the energy is lost, as printed.
- **Torque = power ÷ speed.** At hover the compressor takes all 1,810 hp of the gas-generator turbine; the power turbine gives 911 hp (Table B.1: 911.1).

Eqs. 3–41, pdf pp.22–25; constants Table A.1, pdf p.55; trim Table B.1, pdf p.67

Volume dynamics: mass storage only

$$P3 = K_{V3} \int T3 \cdot (WA3 - WA3_{bl} - WA31)\, dt \quad (42)$$

$$P41 = K_{V41} \int T41 \cdot (WA31 + W_f - W41)\, dt \quad (43)$$

$$P45 = K_{V45} \int T45 \cdot (W41 - W45 + B_3 K_{bl} WA2)\, dt \quad (44)$$

- “Conservation of mass is applied to determine intervolume pressure dynamics for stations 3, 4.1, and 4.5” [pdf p.25].
- Each pressure integrates the net mass flow into its volume, scaled by the gas temperature there. $K_{V3} = 0.97$, $K_{V41} = 6.17$, $K_{V45} = 13.63$: empirical constants; the report gives no volumes.
- **Limitation:** there is no energy equation for the volumes. Their temperature is set by the component upstream, so a volume stores mass but not energy.
- In the real-time model these three are set to zero and solved, not integrated.

Eqs. 42–44, pdf p.25; Eq. 73, pdf p.36; Table A.1, pdf p.55

Spool dynamics

$$NG = (60/2\pi) \int (Q_{GT} - Q_C) / J_{GT}\, dt \qquad (45)$$

$$NP = (60/2\pi) \int (Q_{PT} - Q_{req}) / (J_{PT} + J_{load})\, dt \qquad (46, 47)$$

- **Gas generator:** turbine torque minus compressor torque; $J_{GT} = 0.0445$ ft·lbf·s².
- **Power turbine:** turbine torque minus the load; $J_{PT} = 0.062$ ft·lbf·s². The turbine torque carries a damping term proportional to $NP - NP_{des}$.
- The load torque Q_{req} and the load inertia J_{load} come from the helicopter simulation, which the report does not contain.
  - We recovered both from Appendix B's own matrices: $J_{load} = 9.25 \times J_{PT}$, consistent to 0.2 % across the three flight conditions (Part 4 shows how).
- Trims and the open-loop transients hold NP, as the report does [pdf pp.39, 67]; the closed loop integrates it.

Eqs. 45–47 and Table A.1, pdf pp.25, 55; Appendix B, pdf pp.68–76

Heat soakage at station 4.1

"There is a transient energy transfer which is caused by the metal mass of the turbine absorbing heat from the hot gas. This is especially significant at station 4.1, immediately downstream of the combustor." [pdf p.25]

- Metal: $c_{pm}\, M\, dT_m/dt = h\, A_m\, (T_{gi} - T_m)$ (48)
- Gas: $c_{pg}\, W_g\, (T_{gi} - T_{go}) = c_{pm}\, M\, dT_m/dt$ (49)
- The time constants change with T41, airflow and speed (Eqs. 51–53, $TC_{T41} = 0.29$ and the function f_{hs}, Fig. A11).
- We integrate the **metal temperature** (Eqs. 48–49). The report's lead–lag form (Eq. 50) assumes constant coefficients, and goes stale on a fuel step.
- **Limits Ballin states:** the same constants for heating and cooling; no heat sink at station 4.5.

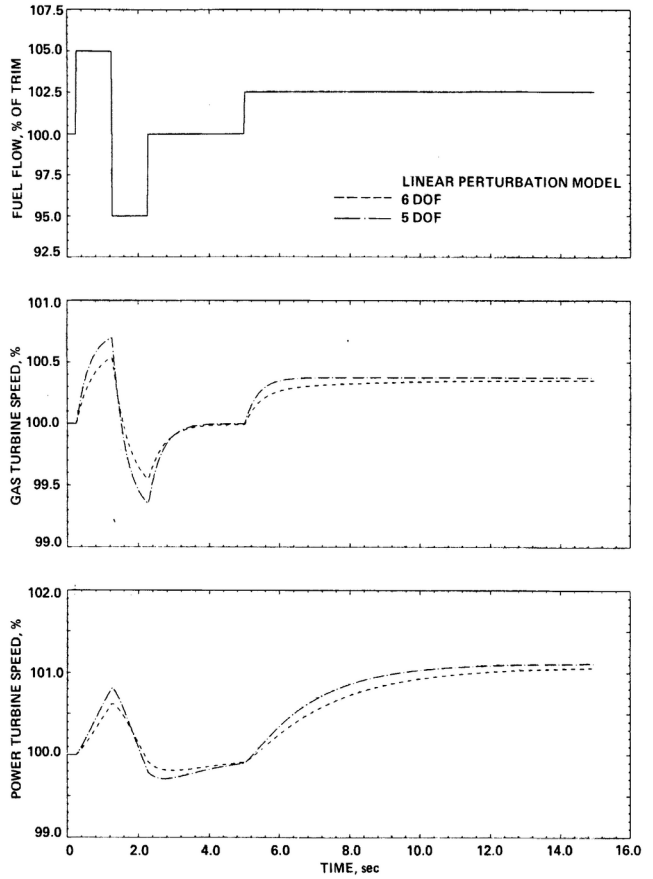
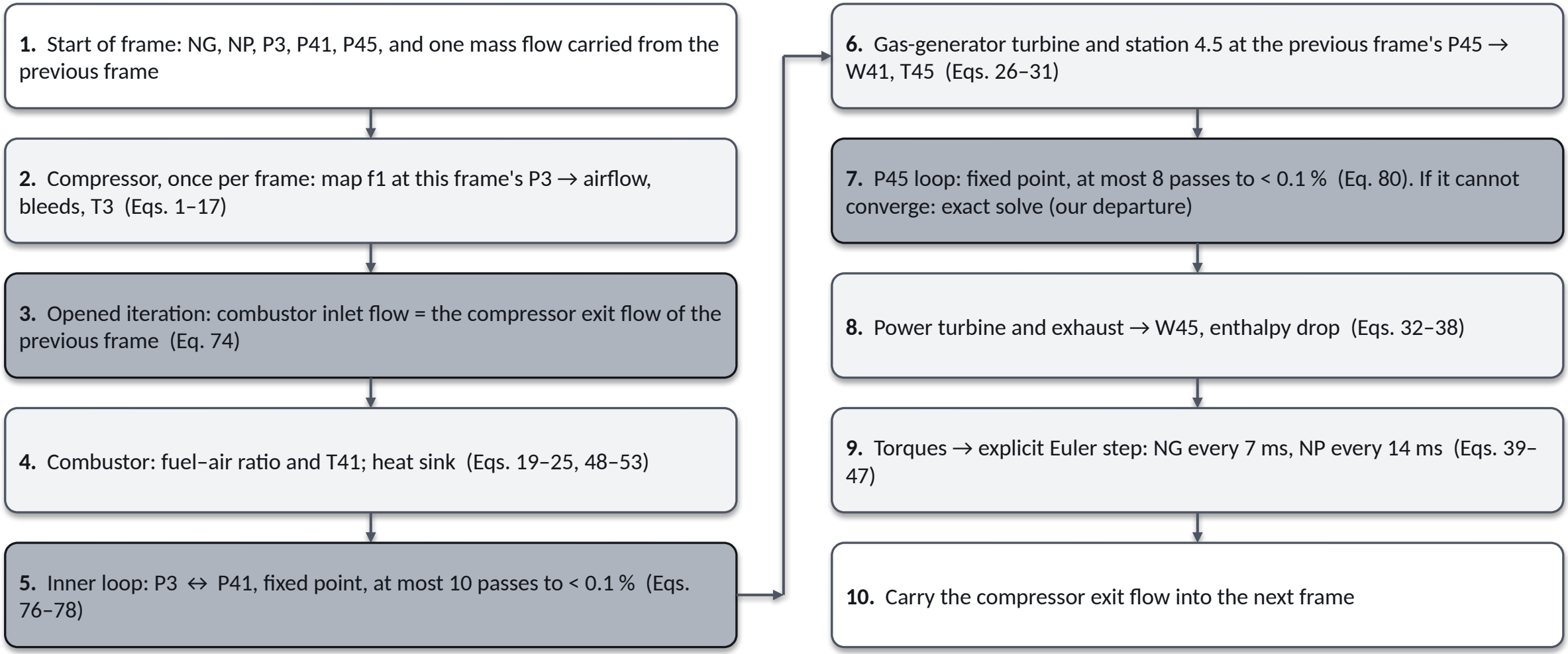


Figure 5: Linear model responses at a high-power flight condition with and without station 4.1 heat-sink model.

Ballin's Figure 5: linear responses with (6-DOF) and without (5-DOF) the station 4.1 heat sink.

Eqs. 48–53, pdf pp.25–26; Ballin, Figure 5, pdf p.34; limits pdf pp.47, 50

One real-time frame, step by step



Our flowchart, from Eqs. 69–80 and pdf pp.35–37: the report prints none. Dark grey: the real-time approximations. Step 10 loops back to step 1.

What the real-time approximation costs

- The opened loop leaves a one-frame lag in the combustor inlet flow, so the answer depends on the step size: *"Because of the opened compressor mass-flow iteration, however, response is dependent on time step."* [pdf p.38]
- Ballin capped the step at **10 ms**, for at most 0.1 % error between time steps [pdf p.38]. He ran the engine every **7 ms** and NP every **14 ms**, with the rotor [pdf p.47].
- Why fixed-point iteration rather than Newton: *"Because multiple passes are required to determine elements of the Jacobian for quadratic convergence methods, they were found to involve more computation than linearly convergent methods."* [pdf p.36]
- The report names no integrator. We use **explicit Euler**, consistent with the "no iteration between time steps" of earlier real-time models [pdf p.35].
- **On modern hardware, speed is no longer the limit.** Ballin's engine had "only a small percentage of the available computation time" on a 1988 CDC 7600 [pdf pp.35–36]. Today one frame takes well under a millisecond on any ordinary processor, even in plain Python. We keep his 7 ms because the response depends on the step.

pdf pp.35–38, 47

Fuel control, part 1: the HMU (hydromechanical unit)

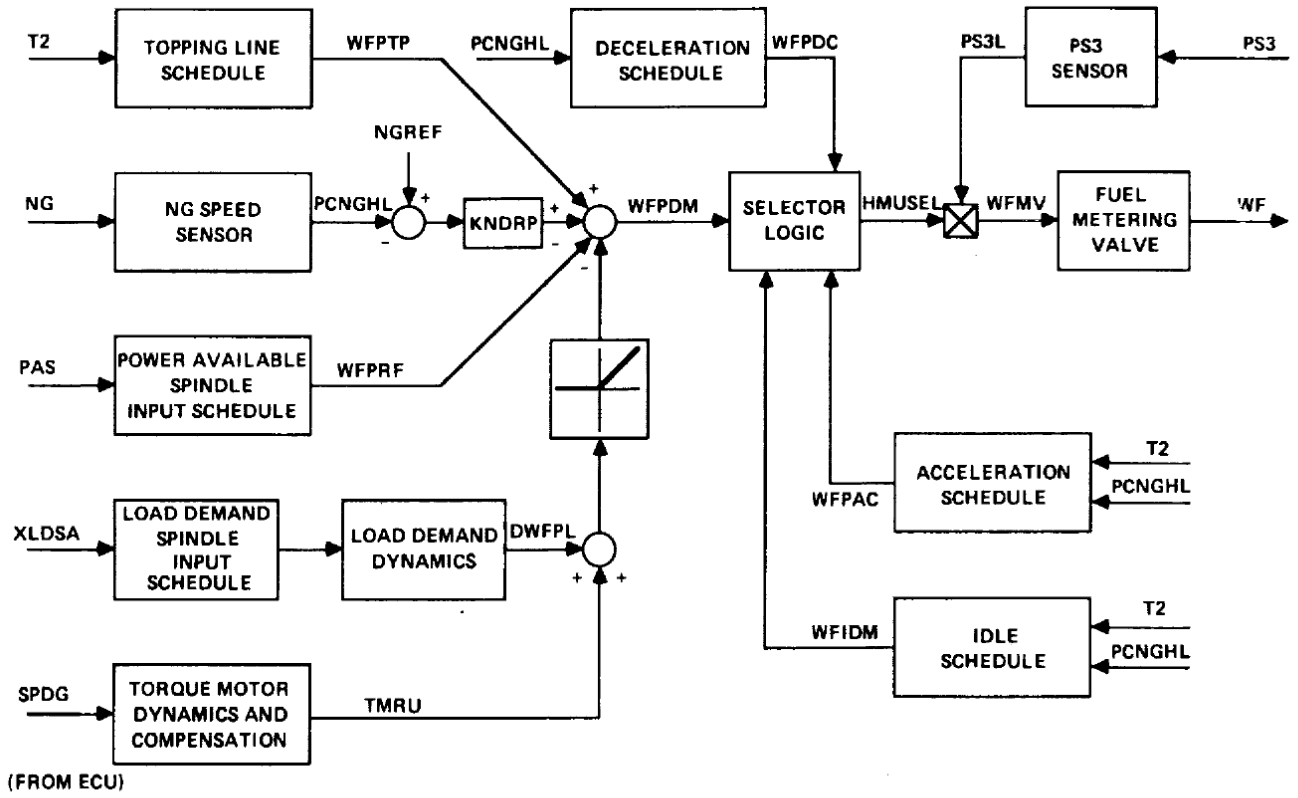


Figure C9.- T700 hydromechanical control unit (HMU).

- The **primary loop**: governs NG on a droop line, applies the idle, acceleration and deceleration limits, and meters fuel. It runs alone if the electrical trim is null.
- Droop line, read sign by sign off the diagram:

  WFPDM = WFPTP + KNDRP·(NGREF − PCNGHL)

  − WFPRF − max(DWFPL + TMRU, 0)
- Limit selection is a fixed cascade (Fig. C18):

  HMUSEL = max(min(max(WFPDM, WFIDM), WFPAC), WFPDC)
- Everything above is a fuel-flow *parameter*, Wf/Ps3. It becomes fuel flow at the metering valve: WFMV = HMUSEL × PS3L.

Ballin, Figure C9, pdf p.89; Figures C18–C19, pdf p.92

Fuel control, part 2: the ECU (electrical control unit)

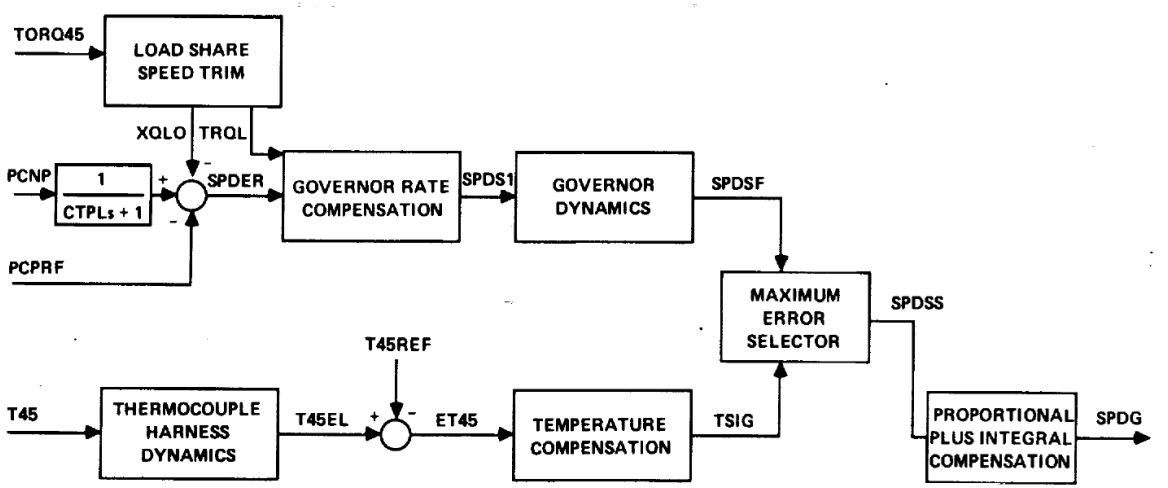


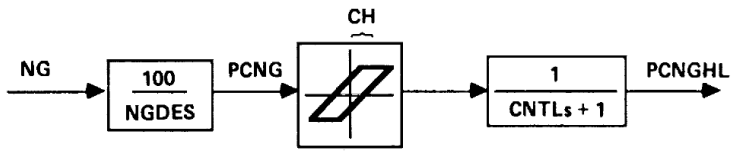
Figure C1.- T700 electrical control unit (ECU).

- Governs **power-turbine speed NP** and **limits T45**.
- Its only output, **SPDG**, drives a torque motor that trims the HMU's droop line downward: more SPDG, less fuel.

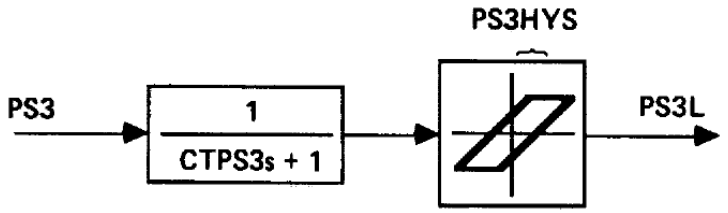
- Speed error: SPDER = PCNP/(CTPL·s + 1) − XQLO − PCPRF
- Temperature error: ET45 = T45EL − T45REF
- Each error is compensated; a **maximum-error selector** passes the larger to the proportional-plus-integral stage, so T45 limiting takes over from speed governing when its error is the bigger.
- Single engine, as drawn: the load-share link (Fig. C2) is open.
- At every Table B.1 trim the sensed T45 sits about 360–570 °R below its limit, so the temperature path is idle in steady flight.

Ballin, Figure C1, pdf p.85; Figures C2–C8, pdf pp.85–88

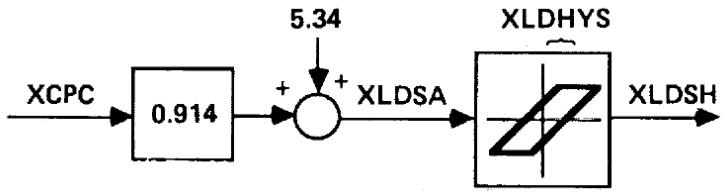
Sensor and actuator dynamics



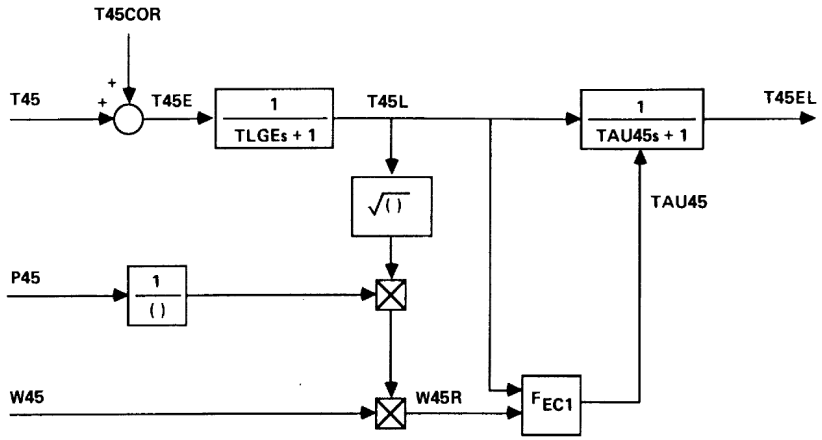
NG speed sensor (C15): 0.05 %NG hysteresis band, then a lag



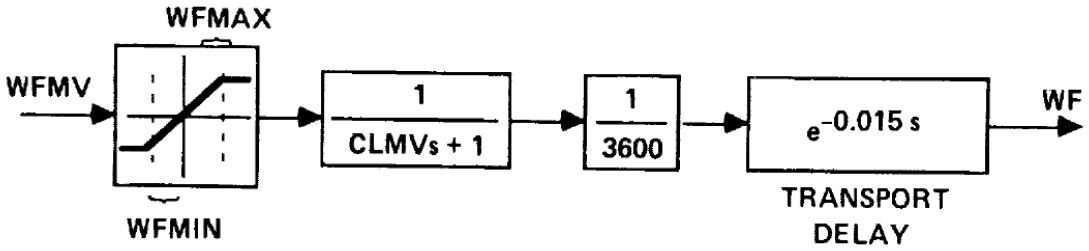
Ps3 sensor (C11): a lag, then a 0.375 psia hysteresis band



Collective rigging (C13): 2.5° hysteresis (UH-60A)



T45 thermocouple harness (C5): two lags, the second one scheduled (F_EC1)



Fuel metering valve (C19): limits, a lag and a 0.015 s transport delay

In all, the fuel control has **22 continuous states** and **4 memory elements** (three hysteresis bands and a transport delay).

Ballin, Figures C5, C11, C13, C15, C19, pdf pp.87–92; Table C.1, pdf pp.83–84

Assumptions and shortcomings

Not in the model

- Energy storage in the gas volumes: mass conservation only
- Tip-clearance effects: the report never mentions them
- Heat soakage anywhere but station 4.1
- Map dynamics: every compressor and turbine map is a steady lookup
- Inlet-guide-vane dynamics [pp.38, 50]
- The engine clutch the report describes [p.19]
- The helicopter: rotor load and inertia come from outside
- Operation above 100 %NG [p.39], or fuel control below flight idle [p.38]

What Ballin says about his own model

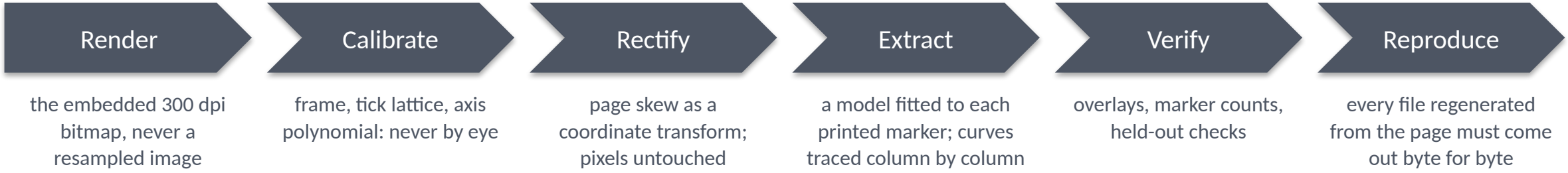
- T41 and T45 at trim slightly low in every validation case [p.47]
- Heat-sink constants the same for heating and cooling [pp.47, 50]
- Damping slightly greater than the test data [p.50]
- Load-demand compensation weaker than the aircraft's [pp.47, 50]
- Response depends on the time step [p.38]
- Not valid at low power-turbine speeds [p.39]

Everything in both lists is preserved as Ballin had it, to match his results as closely as possible.

Ballin's admitted limits: pdf pp.38, 39, 47, 50, 54

From a 1988 scan to numbers

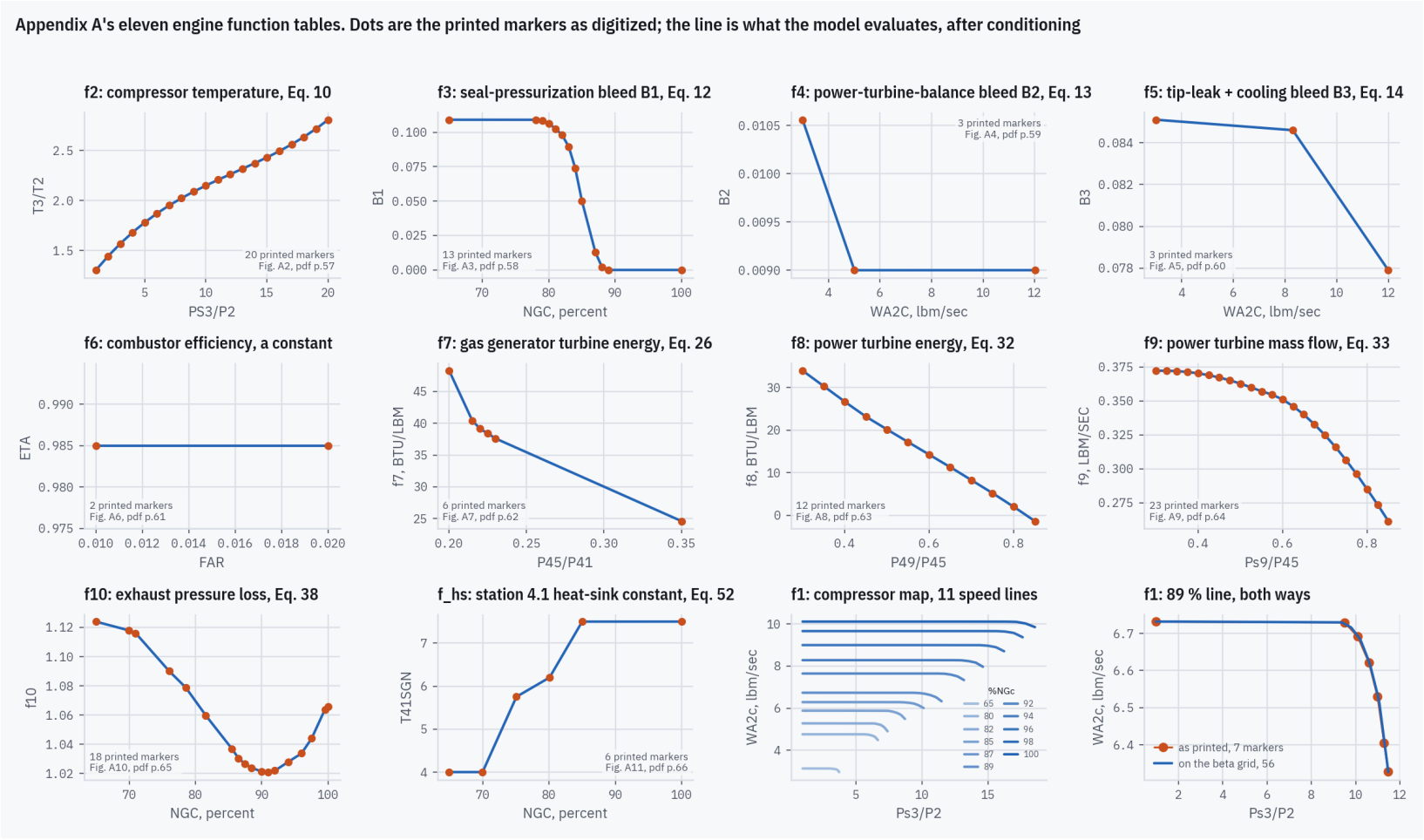
The report prints its 19 function tables only as plots: 11 engine functions (Appendix A) and 8 control schedules (Appendix C). Twelve body figures, 4 to 15, carry the validation references. Recovering them is a measurement, treated as one.



- **The read error is measured, not guessed.** Figures 9 and 10 include the fuel-flow input, whose two levels the caption prints. Digitizing that panel measures the read error directly.

Pipeline: the project's digitizing tools; read-error method: Figures 9–10, pdf pp.45–46

Appendix A's engine functions, as digitized



Orange dots are Ballin's printed markers as digitized; blue lines are what the model evaluates, after conditioning. The compressor map f1 (bottom row, third) has 11 speed lines on our beta grid; the last panel draws its 89 % line two ways.

Figures A1–A11, pdf pp.56–66

How accurately the maps are read

Map	What it gives	Against	Points	Value axis	Argument axis
f1	compressor corrected flow WA2c	PS3/P2, NGc	77	0.05–0.16	0.04–0.45
f2	compressor temperature ratio T3/T2	PS3/P2	20	0.03	0.04
f3	stage-4 seal bleed fraction	NGc	13	0.14	0.12
f4	PT balance bleed fraction	WA2c	3	0.12	0.20
f5	tip leakage and cooling bleed	WA2c	3	0.17	0.16
f6	combustor efficiency	fuel-air ratio	2	0.04 of value	0.11
f7	GG turbine enthalpy drop / θ41	P45/P41	6	0.05	0.04
f8	PT enthalpy drop / θ45	P49/P45	12	0.11	0.07
f9	PT corrected flow	Ps9/P45	23	0.03	0.03
f10	exhaust pressure ratio P49/Ps9	NGc	18	0.11	0.13
f_hs	heat-sink constant	NGc	6	0.12	0.12

1σ read error per printed point, median, in % of each curve's span on that axis; f1 by marker family, from its header table.

- **Measured, not estimated.** Each figure's own redundancy is the ruler: printed frame values, the tick lattice, construction lines, markers printed at known positions.
- **Typical 1σ: about a tenth of a percent of a curve's span.** Small for a value: the trims match Table B.1 to 0.22 % rms.
- **Large for a slope.** A linear model differences two readings close together, across the breakpoints of a piecewise-linear table: Part 5 shows what that costs.
- f1's weakest line is the 65 % one, whose six sloped markers overprint each other.

Each map file's own header (data/maps/*.csv); Figures A1–A11, pdf pp.56–66. Backup slide, hidden in the show

Implementation, in brief

- **Python 3 and NumPy only** in the model; SciPy and Matplotlib only in the analysis tools.
- **One function per component**, in the report's order. Variable names are the report's symbols, and each constant carries its page.
- **Deterministic:** the same input gives bit-identical output, every run.
- **Over 1,500 automated checks.** They include every comparison in this presentation, each with an acceptance bound that cannot be widened silently.
- **Every digitized file** is regenerated from the page images on demand, and must come out identical.
- **Public:** github.com/evrenge/t700-engine-model; every result and figure is in its results/ folder.

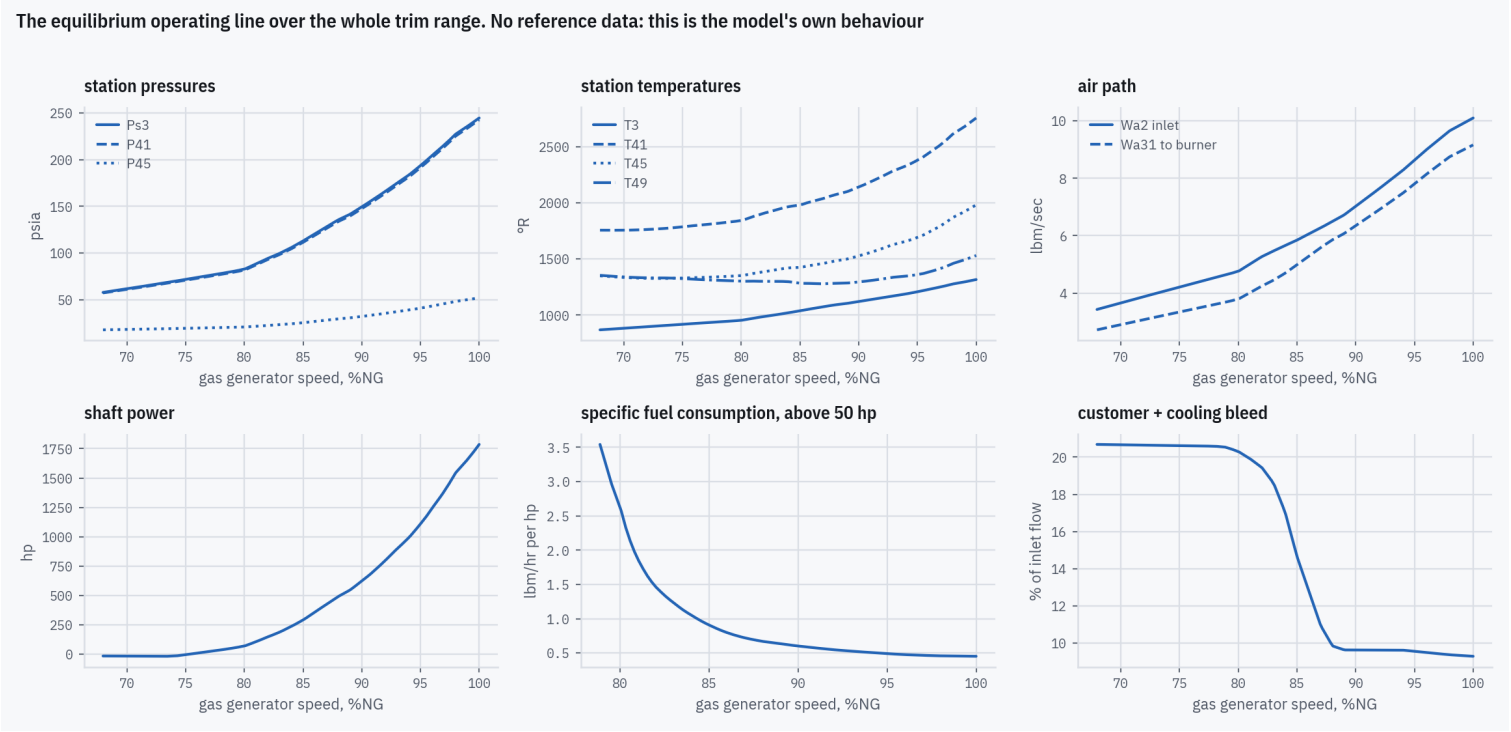
Backup slide, hidden in the show

Five ways the engine is run

Scenario	Model	Given	Solved / output	Compared with
Steady state (trim)	simple nonlinear, all derivatives zero	fuel flow, NP, ambient	NG, P3, P41, P45, shaft torque	Table B.1; Figures 6–8; Table 3
Open-loop transient	real-time, 7 ms frame	fuel flow against time; NP held	everything else, in time	Figures 9–10
Closed loop	real-time + ECU + HMU	load torque, pilot inputs	fuel flow, speeds, the whole state	Table B.1, found on its own
Manoeuvres	closed loop	a load step, collective or speed reference; or Ballin's printed inputs	responses	Figures 11–15, on Ballin's printed inputs
Linear	nonlinear models at a trim, ±2 % differences	trim state	A and b matrices, eigenvalues	Table 1; Appendix B; Figures 4–5

Steady state: trimming the engine

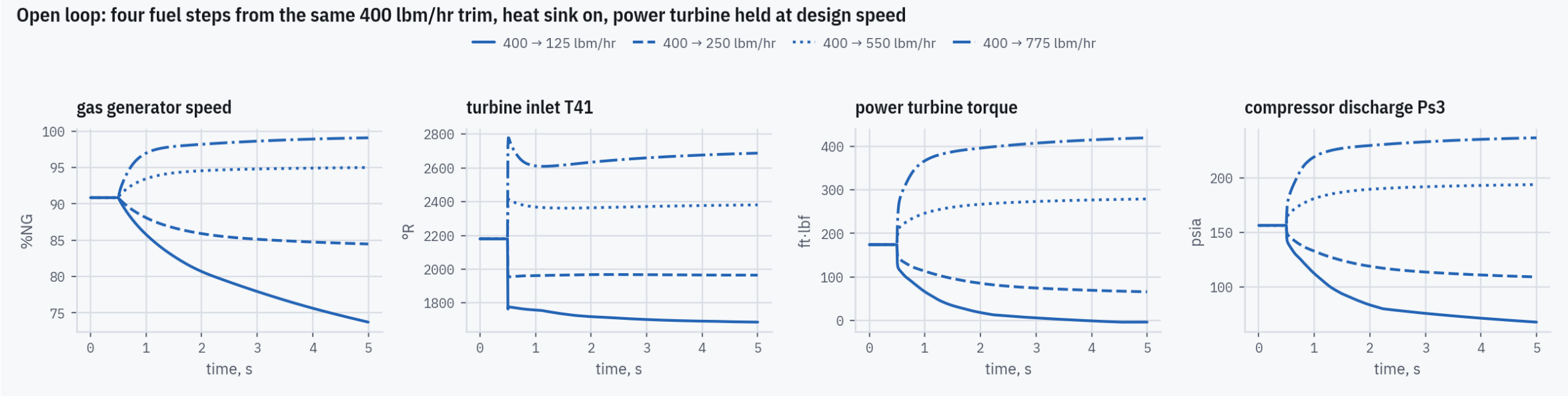
- A trim is an equilibrium of the simple nonlinear model: we solve dNG/dt = dP3/dt = dP41/dt = dP45/dt = 0 for NG, P3, P41, P45 at given fuel flow, NP and ambient.
- NP is held: the load belongs to the helicopter. The shaft torque that balances it is an *output*, and Table B.1 prints it, so it is a check.
- Damped Newton with a finite-difference Jacobian.
- Shaft power goes negative below 75.3 %NG: there the free turbine absorbs power, and the report drops fuel control below flight idle [pdf p.38].



The equilibrium operating line, fuel flow 130–806 lbm/hr, power turbine at design speed. Not compared with anything: this is the model on its own.

pdf pp.38, 67

Open loop: fuel steps

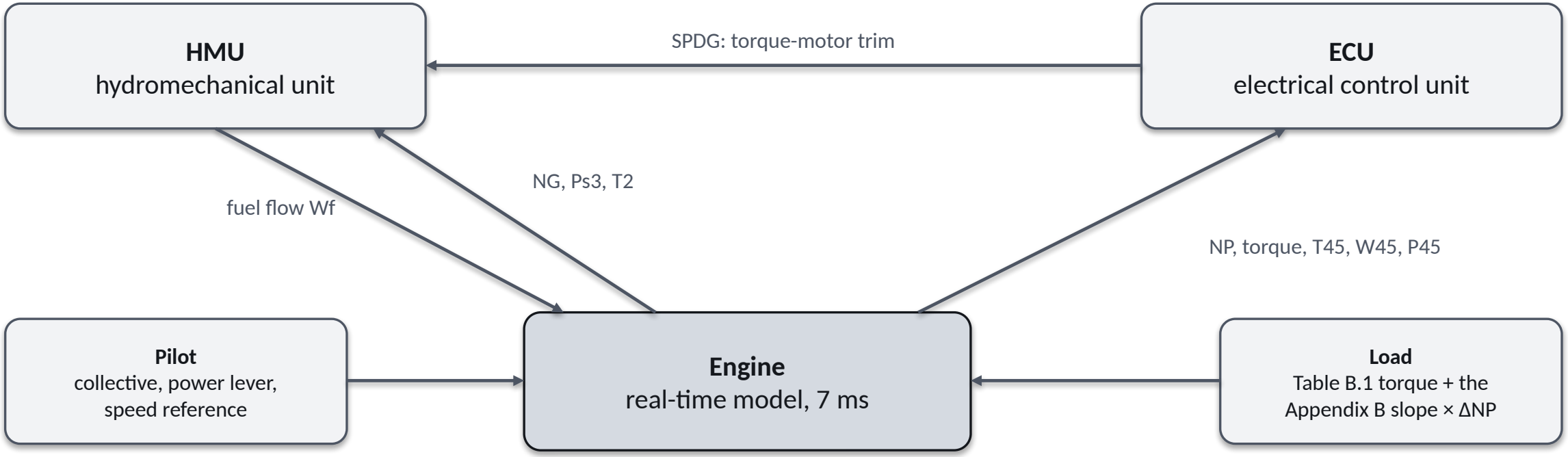


Step from 400 to, lbm/hr	NG start, %	NG end, %	T41 peak, °R	T41 settled, °R	Overshoot, °R
125	90.9	73.7	—	1,687	—
250	90.9	84.5	—	1,965	—
550	90.9	95.0	2,414	2,381	33
775	90.9	99.1	2,778	2,688	91

- As in Figures 9–10: NP held “constant by suppressing the NP integration” [pdf p.39], heat sink on, 7 ms frame.
- The T41 spike: burner temperature moves before the spool can respond; the station 4.1 heat sink damps it. Peak and overshoot apply to the steps up.

pdf p.39; Figures 9–10, pdf pp.45–46

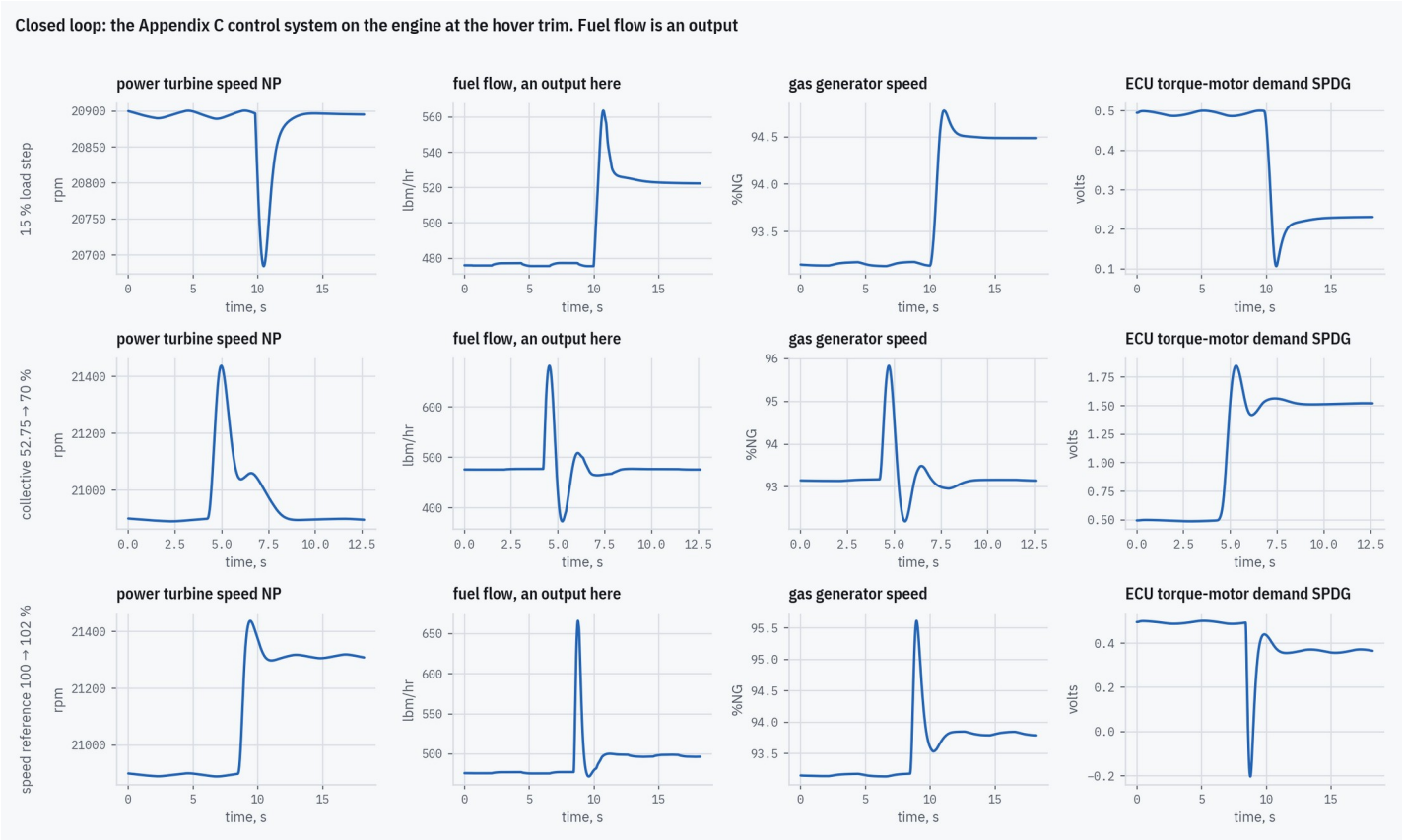
Closed loop: engine and fuel control together



- Fuel flow becomes an *output*. Each control reads the engine's previous frame, as a real-time implementation does.
- **A genuine cross-check:** engine from the report's body and Appendix A, control from Appendix C. Hold Table B.1's printed shaft torque and set the speed reference: the loop finds the printed state on its own, worst 0.25 % over NG, NP, Wf, Ps3 and shaft power at three trims.
- The NG sensor's hysteresis keeps it in a small limit cycle (NP 12.7–16.3 rpm peak to peak), so closed-loop values are averaged over whole cycles.
- **The load:** Table B.1's torque plus the Appendix B slope times the NP change, on the Appendix B inertia, the same two numbers as the linear models (slides 32–33).

Appendix C, pdf pp.77–101; Table B.1, pdf p.67

Closed-loop manoeuvres



Manoeuvre	NP range, rpm	NP settled	Wf range, lbm/hr	Wf settled
15 % load step	20,685–20,901	20,896	476–564	522
collective 52.75 → 70 %	20,890–21,438	20,898	373–682	476
speed reference 100 → 102 %	20,889–21,438	21,315	472–666	497

- Load: Table B.1's torque plus the Appendix B slope times (NP − 20,895 rpm), with the Appendix B inertia.
- The collective slam exercises only the load-demand feedforward: our load depends on speed alone, so the control anticipates a load that never arrives and NP rises rather than dipping.
- The report's own manoeuvres (Figures 11–15) fly the engine inside the Gen Hel helicopter simulation, which is not in the report. Figures 11–15 are compared instead by running our closed loop on what they print (Part 5).

Our runs at the hover trim; the report's Figures 11–15 (pdf pp.48–53) are compared in Part 5

Linearization: how we built our own linear models

- Ballin: “plus or minus two percent of equilibruim [*sic*] conditions” [pdf p.28]. We take central differences of the nonlinear model at a trim: A = ∂ẋ/∂x, b = ∂ẋ/∂Wf.
- **Three trims** (Table B.1; sea level, UH-60A at 16,825 lbm, NP 20,895 rpm): hover, 911.1 hp; 80 kt level flight, 552.6 hp; 80 kt with 1000 ft/min descent, 302.6 hp.
- **Each variant is made the way Ballin made his** [pdf pp.26–28]: 5-DOF from the simple nonlinear model; 2-DOF from a model with the pressures in balance; reduced 5-DOF by setting the 5-DOF's pressure derivatives to zero.
- **Modes are named by the state that dominates them**, not by sorting: at descent the two fastest modes lie a few percent apart, and sorting would pair them by size, not by physics.
- How each cell is calculated, and how the NP row gets the helicopter load: the next three slides.

	Heat sink off	Heat sink on
Full volume dynamics	5-DOF B2, B4, B6	6-DOF B8, B10, B12
Pressures in balance	2-DOF B1, B3, B5	3-DOF B7, B9, B11
Order reduction	reduced 5-DOF Table 1 only	—

B1–B12: Appendix B's figures, each printing an A matrix and b vector. 6-DOF adds T41 as a state.

pdf pp.26–33, 67; Appendix B, pdf pp.68–76

Linearization, step by step: how each cell is calculated

"Each state or control was then individually perturbed and the effects of its change were determined in the state derivatives at the input of each integrator." [pdf p.28]

- **1 · Trim.** Find the steady state at the flight condition: every derivative zero. Hover: W_f 476.3 lbm/hr, NG 41,638 rpm, P3 184.3, P41 174.1, P45 37.40 psia; NP held at 20,895 rpm.
- **2 · Freeze.** No time stepping. The model only evaluates its five state derivatives: dNG/dt, dNP/dt, dP3/dt, dP41/dt, dP45/dt (Eqs. 42–47).
- **3 · Push one quantity ±2 %**, everything else held at trim, and evaluate the five derivatives at +2 % and at −2 %.
- **4 · Divide.** Cell = (derivative at +2 % − derivative at −2 %) ÷ (4 % of the trim value). One push fills one column of A; pushing W_f fills b.
  - Example: A(NG,NG) = (−3,800 − 4,224) ÷ 1,666 = −4.82 /s.

Worked example, hover: push NG by ±2 % (±833 rpm)

Row (derivative)	NG − 2 % 40,805 rpm	NG + 2 % 42,471 rpm	Cell = change ÷ 1,666 rpm	Ballin B2
dNG/dt, rpm/s	+4,224	−3,800	A(NG,NG) = −4.82	−4.474
dNP/dt, rpm/s	+14.8	−19.5	A(NP,NG) = −0.0206	−0.02444
dP3/dt, psia/s	−696	+658	A(P3,NG) = 0.813	0.7524
dP41/dt, psia/s	0	0	A(P41,NG) = 0	0
dP45/dt, psia/s	−989	+837	A(P45,NG) = 1.10	1.063

- **NG row**, the gas-generator torque balance (Eq. 45): dNG/dt = (60/2π)(Q_{GT} − Q_C) / J_{GT}. At NG − 2 %: 233.0 − 213.3 = +19.7 ft·lbf; at + 2 %: 223.9 − 241.6 = −17.7 ft·lbf.
- **P3, P41, P45 rows**, the mass balance of each volume (Eqs. 42–44).
- **NP row**, the power-turbine torque balance (Eq. 47). It contains the helicopter load: next slide.

A cell's unit: the row's rate per unit of the column, e.g. (rpm/s) per rpm = 1/s. Six pushes, twelve evaluations: the whole 5-DOF A and b.

Method pdf p.28; Eqs. 42–47 pdf p.25; Table B.1 pdf p.67; Ballin's Figure B2 (5-DOF, hover) pdf p.68

The NP row: where the helicopter load comes in

$$dNP/dt = (60/2\pi) \cdot (Q_{PT} - Q_{req}) / J, \qquad J = J_{PT} + J_{load} \tag{46, 47}$$

$$\Delta Q_{req} = (\partial Q_{req}/\partial NP) \cdot \Delta NP \; + \; (\partial Q_{req}/\partial(dNP/dt)) \cdot \Delta(dNP/dt) \tag{55}$$

	NG	NP	P3	P41	P45	W_f (b)
dNG/dt	engine	0	engine	engine	engine	engine
dNP/dt	engine ÷ J	(engine − load) ÷ J	engine ÷ J	engine ÷ J	engine ÷ J	engine ÷ J
dP3/dt	engine	0	engine	engine	0	0
dP41/dt	0	0	engine	engine	0	engine
dP45/dt	engine	0	engine	engine	engine	engine

Ballin's 5-DOF matrix at hover, zeros as printed. Light grey: engine only. Darker grey: needs the load.

Nothing else depends on NP: its column is zero except the diagonal, so the other four rows, and their eigenvalues, never see the load. *"The power turbine state is completely decoupled from the other states. The neglected dynamics of the simplified load representation therefore have no effect on the gas-generator turbine or volume dynamics."* [pdf p.29]

Push	What changes	NP-row cell
NG, P3, P41, P45	Q_{PT} only: the rotor does not feel them	$(60/2\pi) \cdot \Delta Q_{PT}/\Delta x \div J$
W_f (column b)	Q_{PT} only	$(60/2\pi) \cdot \Delta Q_{PT}/\Delta W_f \div J$
NP itself	Q_{PT} and Q_{req}	$(60/2\pi) \cdot (\Delta Q_{PT}/\Delta NP - \partial Q_{req}/\partial NP) \div J$

- Only *changes* of Q_{req} enter, never its value. At trim $Q_{req} = Q_{PT}$: 228.9 ft·lbf at hover; Table B.1 prints 229.0 ("Engine torque (ref. shaft)").
- So the helicopter supplies just **two numbers**, and the report prints neither: the inertia **J** and the load slope $\partial Q_{req}/\partial NP$. Next slide: how we found them.

Eqs. 46–47, 55, 59–62, pdf pp.25, 28–29; Table B.1, pdf p.67; Figure B2, pdf p.68

Finding the two load numbers in Appendix B

1 · The inertia J, from the fuel cell b(NP)

Push W_f ±2 % (466.8–485.8 lbm/hr): Q_{PT} goes 227.78 → 229.96 ft·lbf.

With J_{PT} alone: b(NP) = 9.549 × 410.8 ÷ 0.062 = 63,275.

Ballin prints 6,174: 10.25 times smaller, and the same at all three trims, so it is an inertia:

$J = 10.25 \times J_{PT} = 0.636$ ft·lbf·s²

2 · Check: the other four NP cells

With that J, nothing fitted (ours vs Ballin, hover):

A(NP,NG): −0.0206 vs −0.02444

A(NP,P3): −43.8 vs −41.28

A(NP,P41): 35.6 vs 33.02

A(NP,P45): 229 vs 232.1

At all three trims: 10 of 12 within 8 %, worst 16 %.

3 · The load slope, from the NP diagonal

Push NP ±2 % (20,477–21,313 rpm): Q_{PT} goes 236.5 → 221.4 ft·lbf, −0.0181 ft·lbf/rpm.

Engine alone: A(NP,NP) = −0.272 /s. Ballin prints −0.565.

The rotor supplies the rest:

$\partial Q_{req}/\partial NP = -0.0181 + 0.565 \times 0.636 \div 9.549$

= 0.0195 ft·lbf/rpm

	Hover	Level 80 kt	Descent 80 kt
b(NP) with J_{PT} alone ÷ Ballin's	10.249	10.260	10.237
A(NP,NP), engine alone, 1/s	−0.272	−0.208	−0.162
A(NP,NP), printed, 1/s	−0.565	−0.4461	−0.3567
Load slope $\partial Q_{req}/\partial NP$, ft·lbf/rpm	0.0195	0.0159	0.0130

What this means for the comparison

The NP eigenvalue is A(NP,NP) itself, so step 3 makes it match by construction: Table 1's 9 NP values are inputs, not tests.
Its other 18 eigenvalues, and every cell outside the NP row, contain no load: they test the engine model alone.

Appendix B, Figures B2, B4, B6, pdf pp.68–70; Eqs. 59–62, pdf pp.28–29; Table A.1 (J_PT), pdf p.55

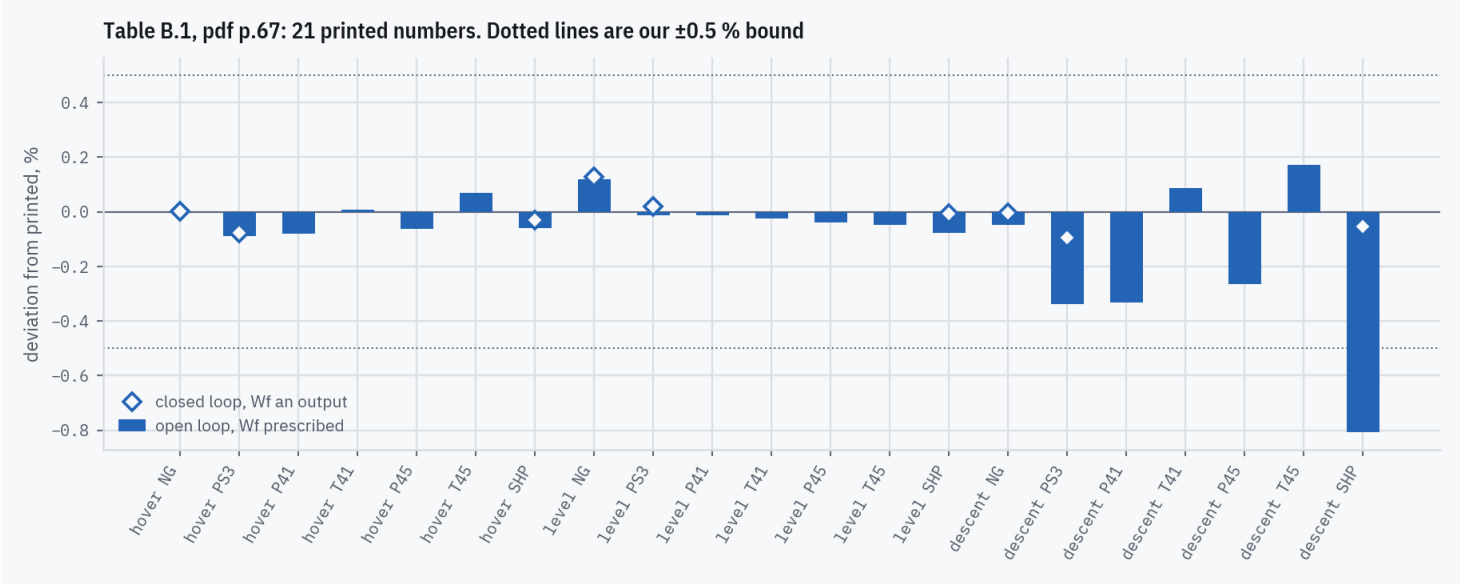
Scorecard: every comparison the report prints

Comparison	What is compared	n	rms	worst
Table B.1, open loop	printed engine state at three trims	21	0.22 %	−0.81 %
Table B.1, closed loop	NG, NP, Wf, Ps3 and shaft power, fuel an output	15	0.08 %	+0.25 %
Figure 6	NG against fuel flow (in %NG points)	31	0.07	−0.23
Figure 7	shaft power against fuel flow (in shp)	25	1.81 shp	+5.16 shp
Figure 8	Ps3 against NG	31	0.22 %	−0.47 %
Table 3 (characterization)	test engine, cases 2–6, against Ballin's model, which ran other maps	30	7.7 %	+13.4 %
Figure 9, fuel step	whole curve, five panels, % of each panel's change	5	0.6–1.4 %	—
Figure 10, fuel chop	whole curve, five panels, % of each panel's change	5	1.4–3.8 %	—
Figures 11–15 (closed loop)	his rotor speed imposed, or our rotor free; % or % of trim	25	0.02–3.2 %	—
Table 1	printed eigenvalues: NG, P3, P41, P45 modes	18	2.37 %	+4.70 %
Table 1, NP mode	supplied from Appendix B (slide 33): not a test	9	—	—
Appendix B, A	printed state-matrix elements, NP diagonal excluded	146	6.5 %	−33.7 %
Appendix B, b	printed fuel-column elements, hover b(NP) excluded	41	6.5 %	+24.3 %
Figures 4–5	linear models in time, rms difference from his curves, % of trim	12	0.005–0.045 %	—

Each comparison has an acceptance bound in the project, set from the scan's read error or today's result, and never widened to pass.

Deviation = (ours − printed) / printed, unless noted

Table B.1: the printed engine state at three trims



Trim	Quantity	Printed	Ours	Dev. %
hover	NG	41,638	41,638	+0.00
hover	T41	2,292	2,292	+0.01
hover	SHP	911.1	910.6	−0.06
level	NG	39,768	39,816	+0.12
level	T41	2,102	2,101	−0.02
level	SHP	552.6	552.2	−0.08
descent	NG	38,072	38,054	−0.05
descent	T41	1,982	1,984	+0.09
descent	SHP	302.6	300.2	−0.81

- 21 printed numbers: NG, Ps3, P41, T41, P45, T45 and shaft power at hover, level flight and descent. Open loop rms 0.22 %, worst −0.81 % (descent shaft power).
- Bars: open loop, fuel prescribed. Diamonds: closed loop, where fuel is an output and the engine must find the printed state by itself.

Table B.1, pdf p.67

Figures 6–8: the steady sweeps

Figure 6, pdf p.40: NG against fuel flow

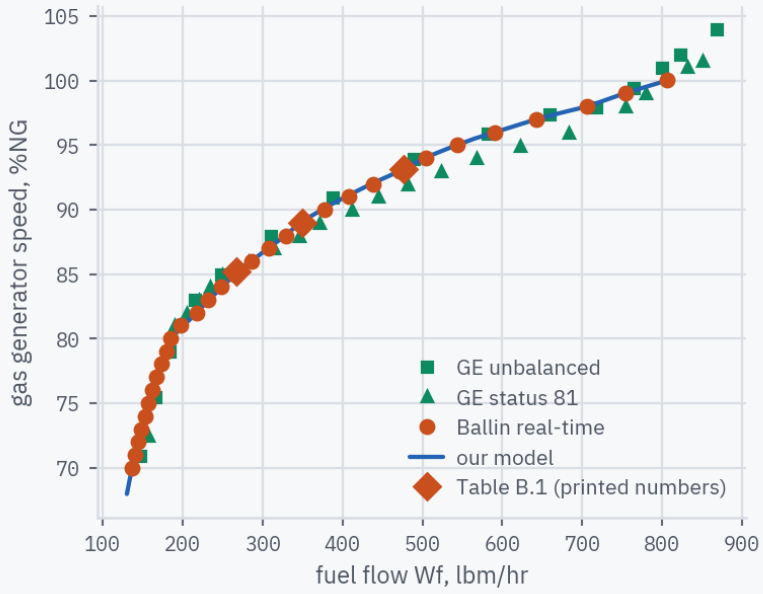


Figure 7, pdf p.41: shaft power against fuel flow

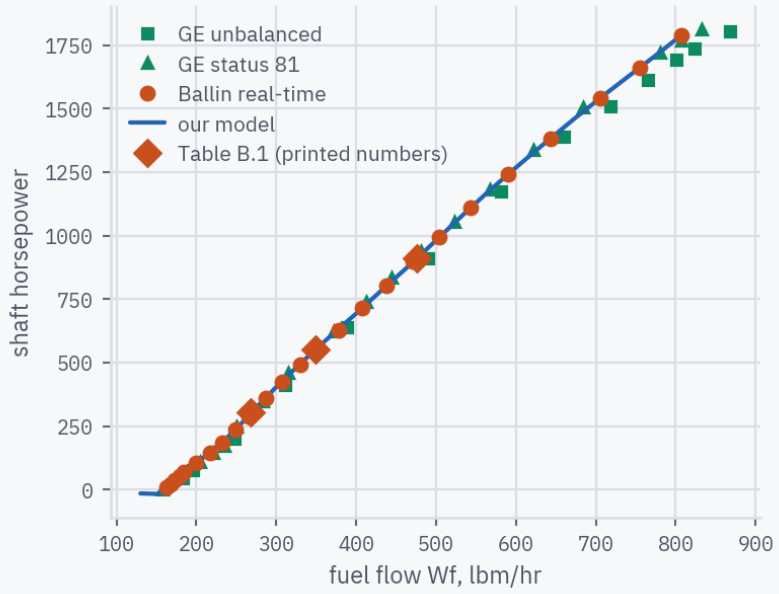


Figure 8, pdf p.42: Ps3 against NG

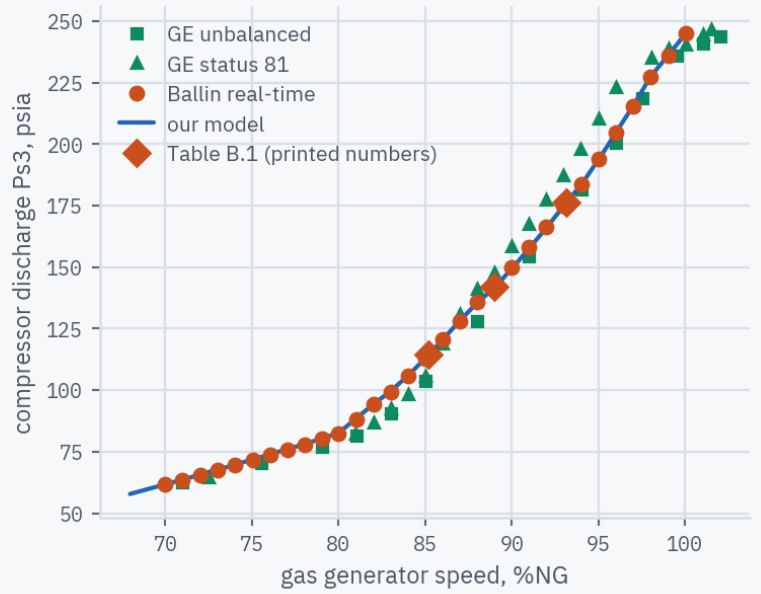


Figure	Points	mean	rms	worst
6: NG vs fuel flow (%NG pts)	31	−0.01	0.07	−0.23
7: shaft power vs fuel flow (shp)	25	+0.96	1.81	+5.16
8: Ps3 vs NG (%)	31	+0.04	0.22	−0.47

- Blue: ours. Orange: Ballin's model, the target (diamonds: Table B.1's printed numbers). Green: GE's two computer models [pdf p.38], for context only (triangles status 81, squares unbalanced torque).
- Agreeing with GE more closely than Ballin does would not be evidence of anything.

Figures 6–8, pdf pp.40–42

Figures 6–8: each point's miss, in units and in percent

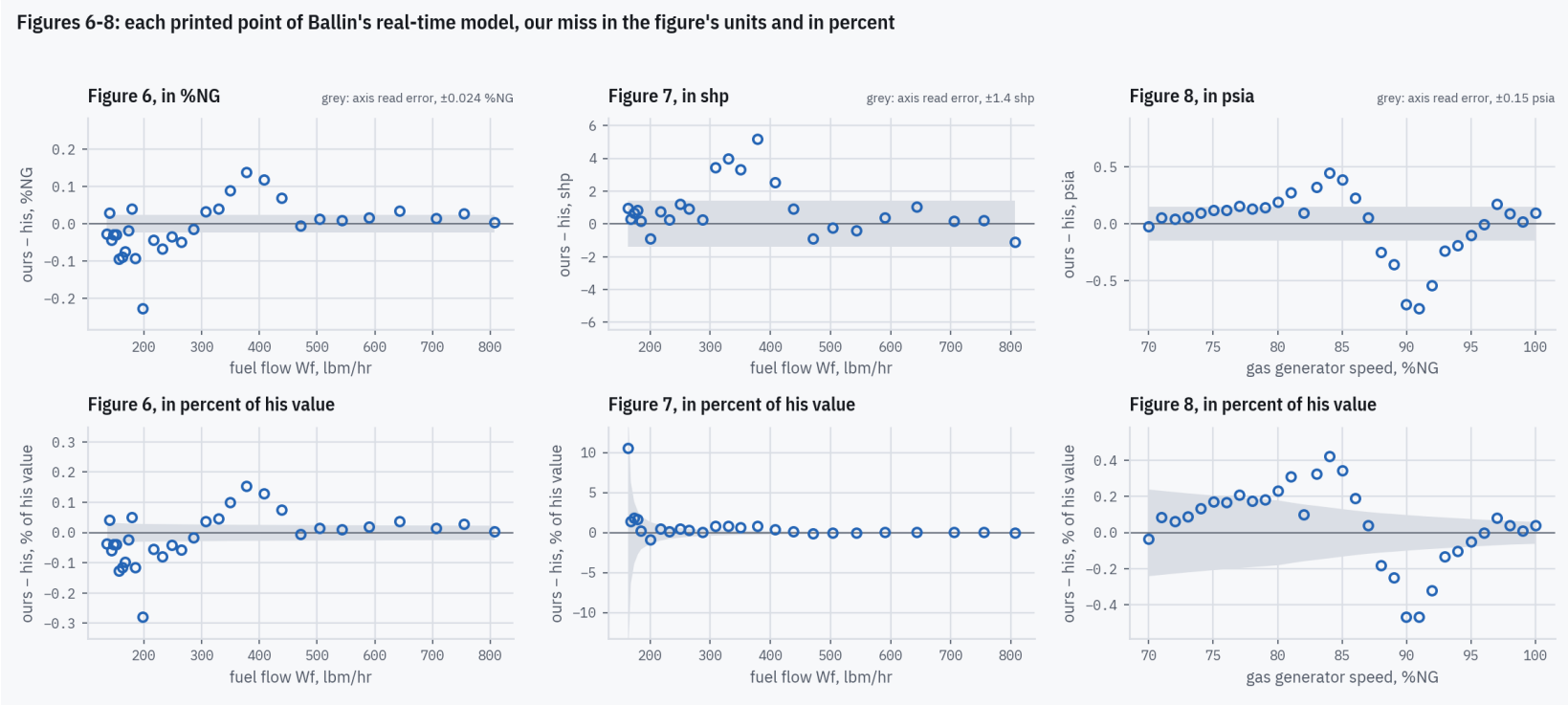


Figure	Units	rms	worst	rms, %	worst, %	Read error
6	%NG (scored)	0.070	−0.227	0.08	−0.28	±0.024 %NG
7	shp (scored)	1.81	+5.16	2.22	+10.57	±1.40 shp
8	psia	0.28	−0.74	0.22 (scored)	−0.47	±0.15 psia

- **Grey:** the axis read error each digitized file publishes. In the figure's units it is constant; in percent it grows as his value falls.
- **Figure 7** falls to 9 shp at low power, where the read error alone is 15 % of the value. Its worst miss in percent, +10.6 %, is +1.0 shp. So Figure 7 is scored in shp.
- **Figure 6** is already a percentage, and **Figure 8**'s Ps3 never falls below 61 psia: either way tells the same story.
- Outside the grey, the gap is larger than the axis reading can explain: up to 5.2 shp on Figure 7 and 0.74 psia on Figure 8.

Figures 6–8, pdf pp.40–42; decision log #83. Backup slide, hidden in the show

The report disagrees with itself

Table B.1 and Figures 6–7 describe the same three operating points. Where they differ, Table B.1 is primary: it has no digitizing error of ours, and Appendix B and Table 1 come from the same trims.

Trim	Quantity	Wf, lbm/hr	Table B.1 vs Ballin's figure	Ours vs Table B.1
hover	NG	476.3	+0.00 %NG pts	+0.00 %NG pts
hover	SHP	476.3	−0.03 %	−0.06 %
level	NG	349.3	−0.02 %NG pts	+0.12 %NG pts
level	SHP	349.3	+0.72 %	−0.08 %
descent	NG	267.7	+0.02 %NG pts	−0.05 %NG pts
descent	SHP	267.7	+1.20 %	−0.81 %

- The widest self-disagreement: 1.20 % (descent shaft power).
- There our gap to Table B.1 is smaller than the report's own.
- Other places the report disagrees with itself are in Part 6.

Table B.1, pdf p.67; Figures 6–7, pdf pp.40–41

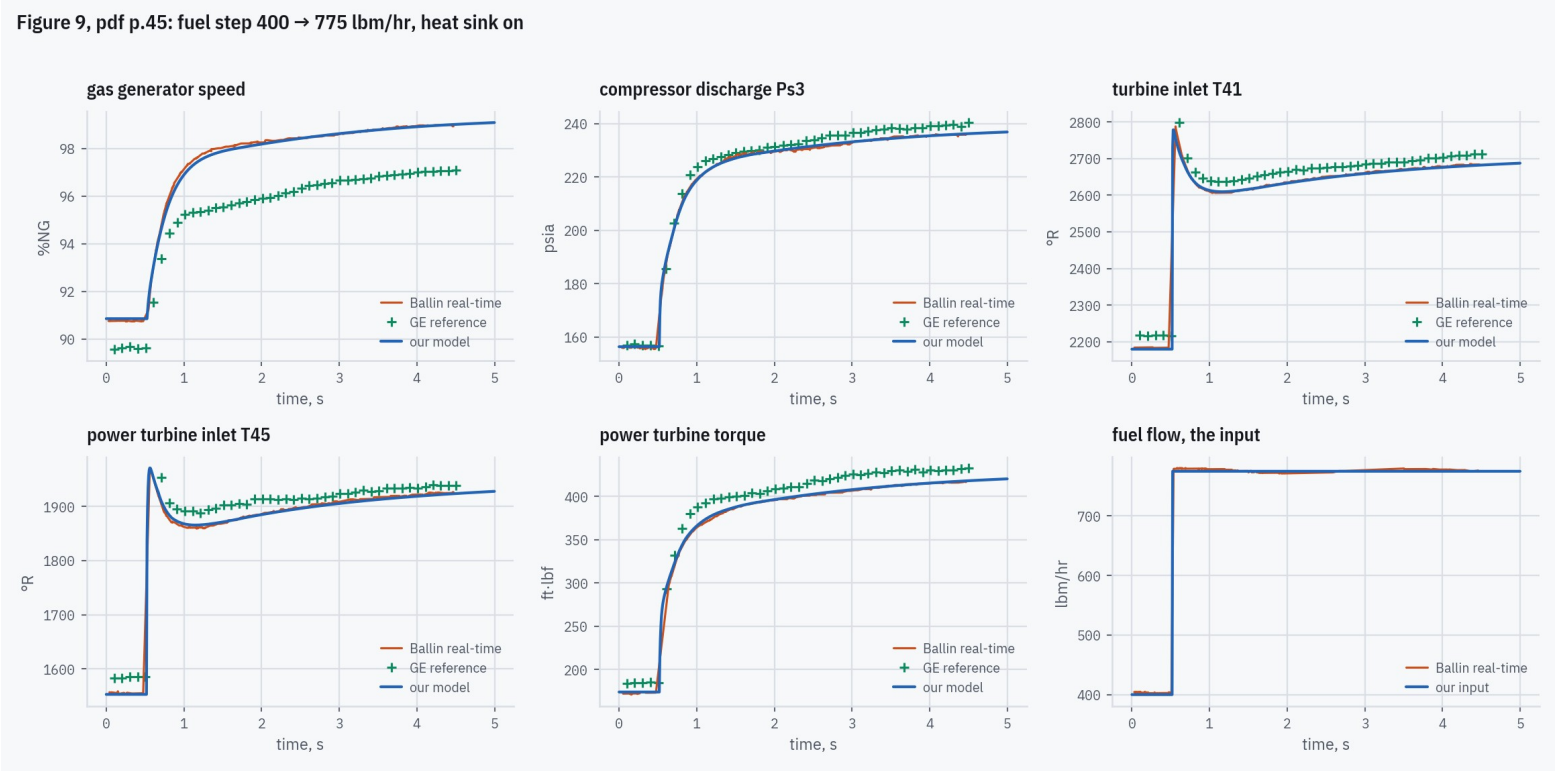
Table 3: a prototype test engine, Ballin's model and ours

Case	Source	NG, %	NP, %	WA2, lbm/s	P3, psia	T3, °R	T45, °R
1	Test engine	65.9	52.6	3.20	58.0	832.0	1,413
	Ballin's model	70.0	83.8	3.59	63.4	855.9	1,372
	Ours	75.2	84.3	4.17	72.6	911.9	1,251
2	Test engine	84.7	95.7	5.16	113.1	1,026	1,577
	Ballin's model	86.4	100.6	5.52	119.0	1,043	1,555
	Ours	86.9	106.7	6.11	130.7	1,071	1,451
3	Test engine	87.7	95.7	6.16	139.0	1,081	1,626
	Ballin's model	88.1	97.8	6.27	141.7	1,084	1,629
	Ours	89.8	104.3	7.05	156.1	1,109	1,509
4	Test engine	90.4	95.7	6.92	161.1	1,127	1,731
	Ballin's model	90.7	97.3	7.00	163.3	1,130	1,734
	Ours	92.6	104.4	7.88	179.9	1,156	1,601
5	Test engine	92.6	95.7	7.66	184.8	1,173	1,838
	Ballin's model	92.8	98.2	7.73	186.4	1,178	1,855
	Ours	95.2	105.5	8.77	206.6	1,207	1,699
6	Test engine	95.9	95.7	8.50	211.9	1,228	1,974
	Ballin's model	96.1	99.4	8.53	213.8	1,233	2,009
	Ours	98.0	104.9	9.48	233.2	1,265	1,860

- Six test-cell points [pdf p.43]: inlet P2 and T2, fuel flow, exhaust P49 and the dynamometer torque. Power-turbine speed floats until its torque meets the load, as Ballin did [pdf p.39].
- **Ballin's column ran the test engine's own maps**, printed nowhere: “compressor-mass-flow and turbine-energy functions derived from this test engine were used in place of the standard functions” [pdf p.39]. Ours runs the standard ones.
- **Cases 2–6, ours against his:** airflow +12.0 %, P3 +10.0 %, T45 −7.5 %, NP +6.6 %, the same sign every time: the standard maps describe a stronger engine than the prototype.
- So the gap measures the prototype against the specification T700, not our model against his. Nothing is tuned toward either column.
- **Not the inlet, not the heat:** with P2 and T2 free, matching his column needs P2 25–28 % below Table 2's and T2 37–43 °R colder (cases 2–5), and still misses by 1.3–1.8 % rms. The cooler T45 is the extra air at the same fuel.

Tables 2–3, pdf pp.43–44; setup pdf p.39

Figure 9: fuel step, 400 → 775 lbm/hr



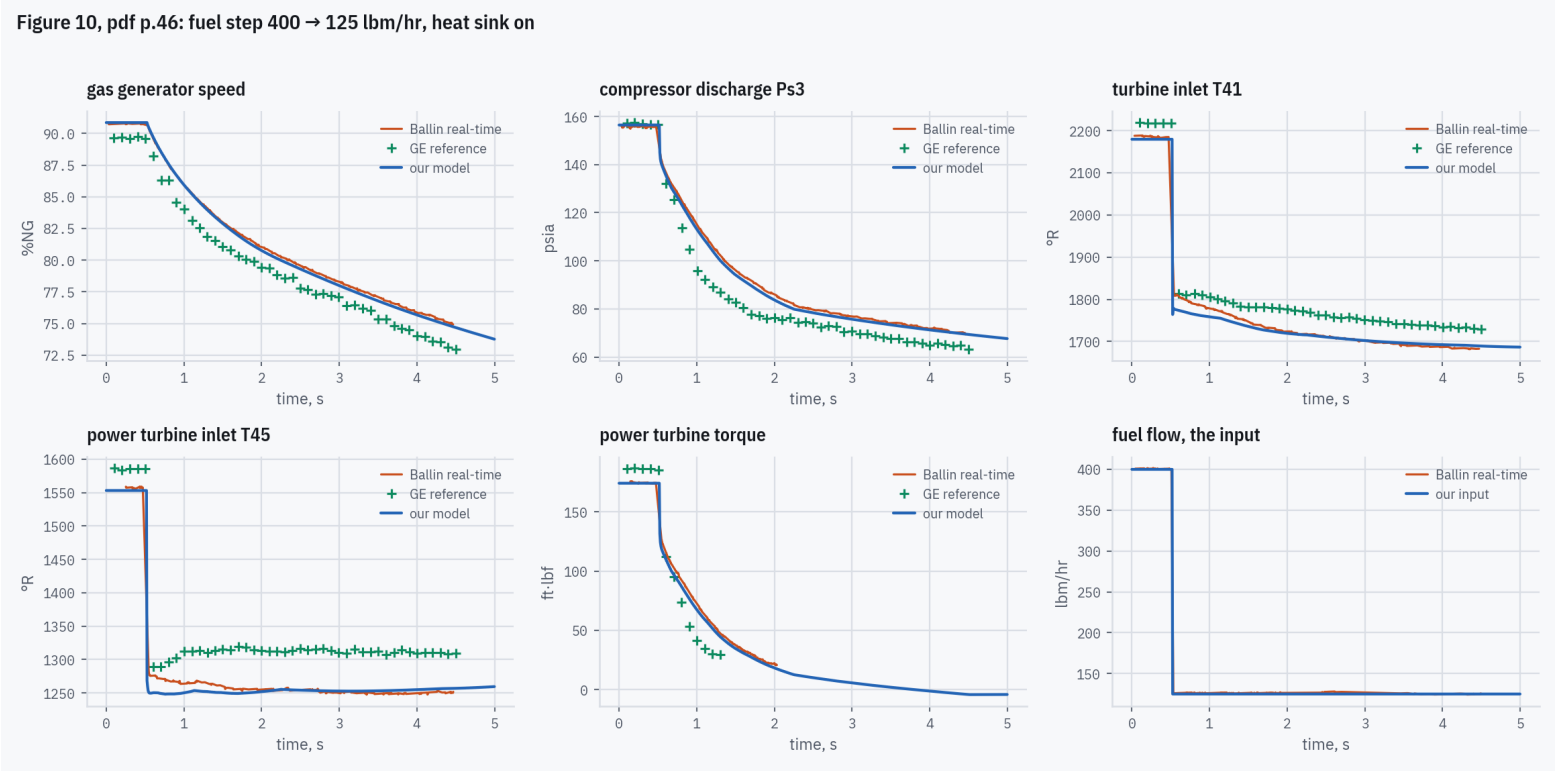
Panel	Ballin	Ours	Dev. %	Shape
NG, %	98.98	98.97	−0.01	1.43
Ps3, psia	235.72	235.79	+0.03	0.71
T41, °R	2,682.0	2,679.0	−0.11	0.67
T45, °R	1,924.4	1,921.3	−0.16	0.82
PT torque	415.98	416.60	+0.15	0.62

- Ballin / Ours / Dev.: the end state, over his final window.
- Shape: rms gap over the whole curve, % of the panel's own change.

- The sixth panel is the input: both levels are printed in the caption, so drawing it shows our model was given Ballin's step, at the instant read off the figure.
- Without the heat sink our T41 spike overshot about three times as far as Ballin's and NG rose too fast. With it the spike is close to his and NG lags slightly (63 % at 0.32 s against 0.28 s).

Figure 9, pdf p.45

Figure 10: fuel chop, 400 → 125 lbm/hr



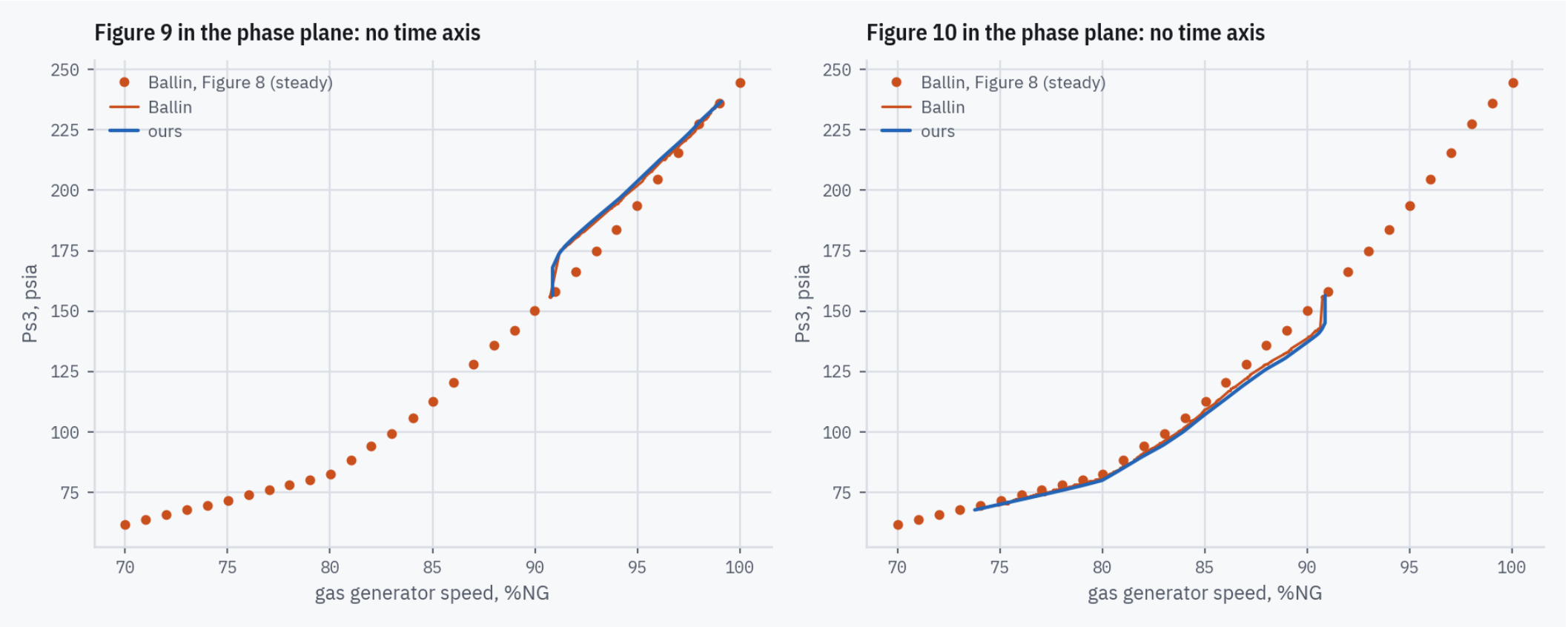
Panel	Ballin	Ours	Dev. %	Shape
NG, %	75.44	75.20	−0.31	1.44
Ps3, psia	70.83	70.38	−0.63	1.68
T41, °R	1,684.0	1,691.0	+0.42	3.80
T45, °R	1,250.6	1,256.0	+0.43	2.90
PT torque	—	—	—	2.39

- Ballin / Ours / Dev.: the end state, over his final window.
- Shape: rms gap over the whole curve, % of the panel's own change.

- Ballin's traces stop at about t = 4.47 s with the speed still falling, so neither has settled; comparisons are made at matched times. His torque line merges with the panel's zero line after about 2 s, so that panel has no end state.
- The chop runs off the printed range of f8, and is where the printed P45 iteration fails (Part 6).

Figure 10, pdf p.46

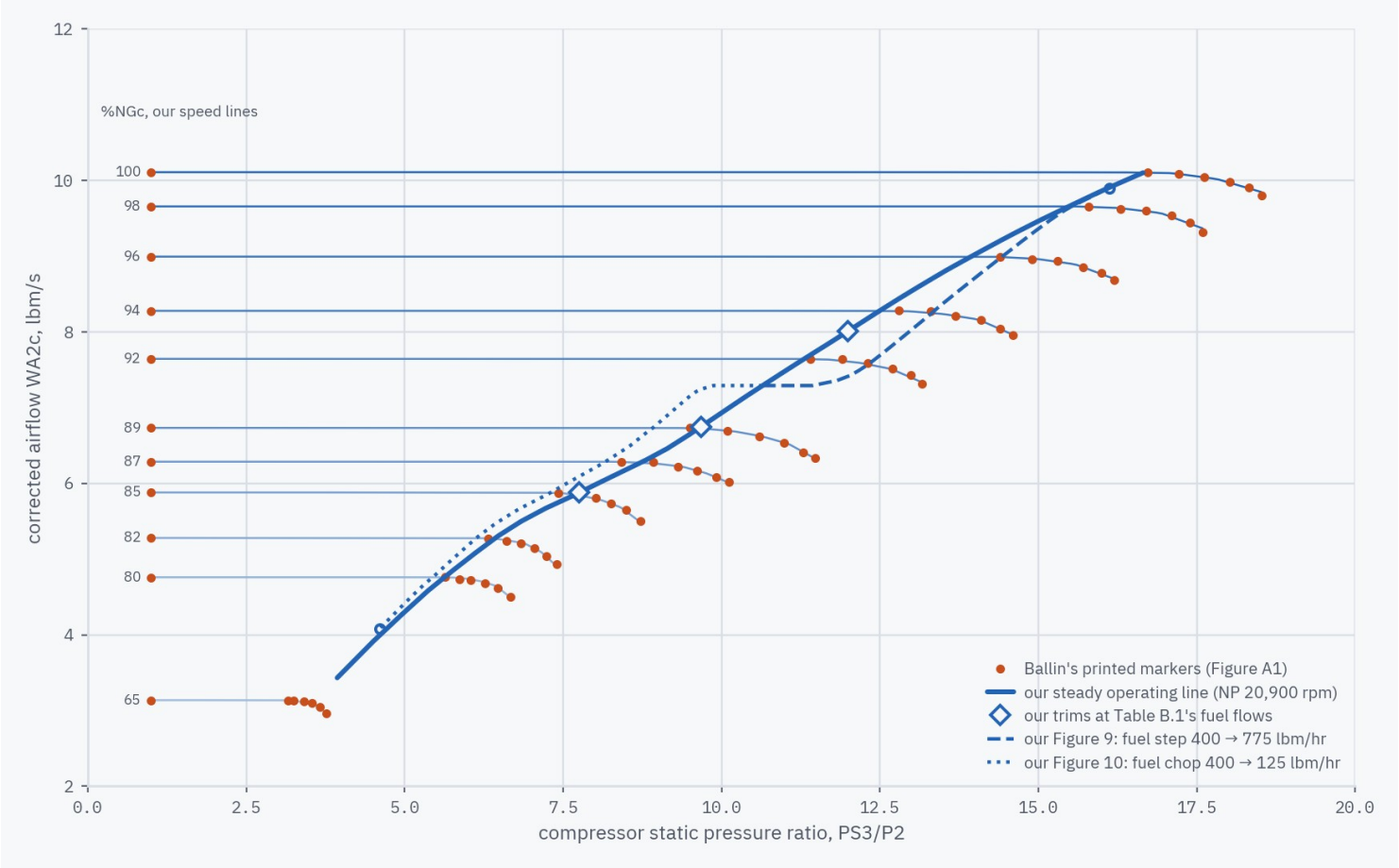
Both transients without the time axis



- Ps3 against NG, so the comparison does not depend on when the step happened. Both transients run below the steady line (orange dots, Figure 8): a choked station 4.1 nozzle passes the same flow at a lower pressure once the temperature falls. Gap to Ballin's path: rms 0.67 % (Figure 9), 1.27 % (Figure 10).

Figures 8–10, pdf pp.42, 45–46

The same runs on Ballin's compressor map



- **Steady operating line** at NP 20,900 rpm, 68–100 %NGc, across the knees of the speed lines. Diamonds: our trims at Table B.1's fuel flows. Orange dots: Ballin's printed map points; everything blue is ours.
- **Fuel step (Figure 9):** the path swings toward the high-pressure-ratio end of the speed lines, as an acceleration does, and stays inside the drawn lines.
- **Fuel chop (Figure 10):** toward lower pressure ratio.
- Neither leaves the compressor map. The turbine maps are where they run off the data: f7, f9 on the step, f8 on the chop.
- Below 80 %NGc Figure A1 has only the 65 % line: the model interpolates across a 15-point gap.
- No surge line: “No modeling of compressor surge … was attempted” in the NASA Lewis model the report builds on [pdf p.20], and Figure A1 draws none.

Figure A1, pdf p.56; Table B.1, pdf p.67; Figures 9–10, pdf pp.45–46; pdf p.20

Figures 11–15: the choices made where the report is silent

Question (the report is silent)	Our choice, confirmed	Why
How to run without Gen Hel	two tests: his rotor speed imposed as our NP, or our rotor free under his torque	“a rigid-shaft drive train model” [pdf p.47]: everything the airframe does reaches the engine through it, except the collective
The flight-test day	a standard day	at his NG our engine burns 1.18–1.23× his fuel, so his day was not standard: fuel and torque are compared in % of trim
Where each run starts	his printed NG and rotor speed	the two engine states his panels print directly
The starting collective	what our fuel control needs to burn our trim fuel	the report prints no collective in our control's unit
Collective, inches to % of maximum (Figures 11, 13)	proportional, 0 in. = 0 %: 8.8 and 9.6 % per inch	our control reads “percent of maximum” [pdf p.81] and his axis starts at 0 in.; the line through both trims would give 13.5 % per inch, and across that range Figure 13's torque rms moves from 1.7 to 10.0 % of trim
Load slope at 55, 87, 90 kt	Appendix B's level 80 kt slope	the nearest printed trim; hover cases take the hover slope
Figure 15's trim power (no NG printed)	his main-rotor torque on our scale	the mean ratio of his torque to ours at Figures 11–14's trims; tail rotor and accessories in proportion
Figure 15's full-down collective	0 % of maximum	“the full-down position” [pdf p.50]; his trace's floor, where it runs on the frame

Decision log #81; pdf pp.47–52, 81, 90

Figure 11: collective input at 90 kt

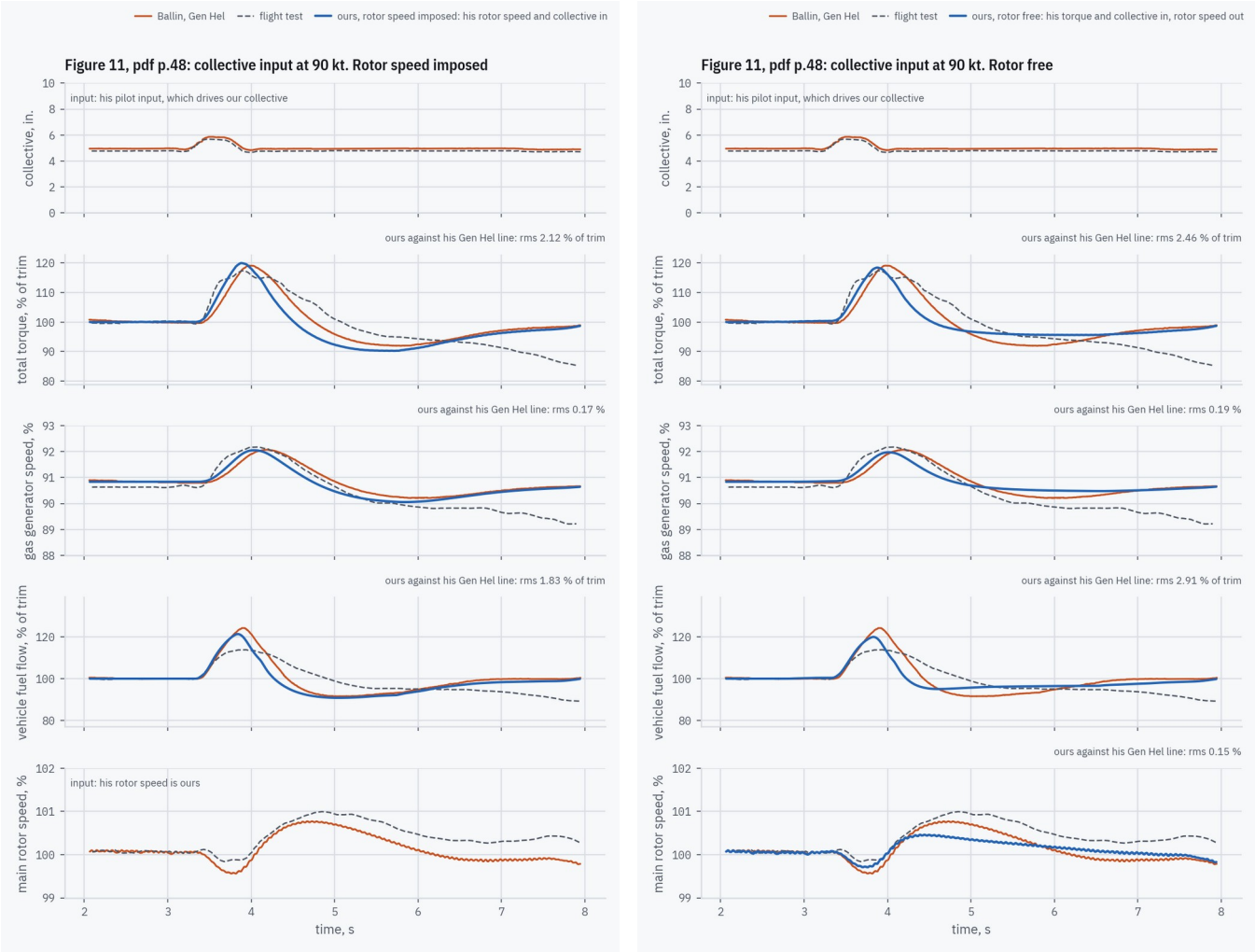


Figure 11, pdf p.48; drive train pdf p.47; decision log #81

Left, rotor speed imposed: his rotor speed is ours; we predict torque, NG and fuel. Right, rotor free: his torque is our rotor's load; we predict the rotor speed too.

Where ours misses his, and why

- **Our fuel turns back sooner.** It peaks at 121 % of trim at 3.84 s, his at 124 % at 3.92 s, and is back under 110 % at 4.09 s against his 4.23 s. Why: in our control his collective reaches the fuel through the HMU's load-demand path (Fig. C17, pdf p.91), so ours follows his collective back down at once. NG and torque follow our fuel.
- **Rotor free:** with less torque as the collective comes back, our rotor overspeeds less, 100.46 % against his 100.77 %. Overspeeding less, our governor then cuts less, so our rotor ends a little above his.

Figure 11: the two tests side by side

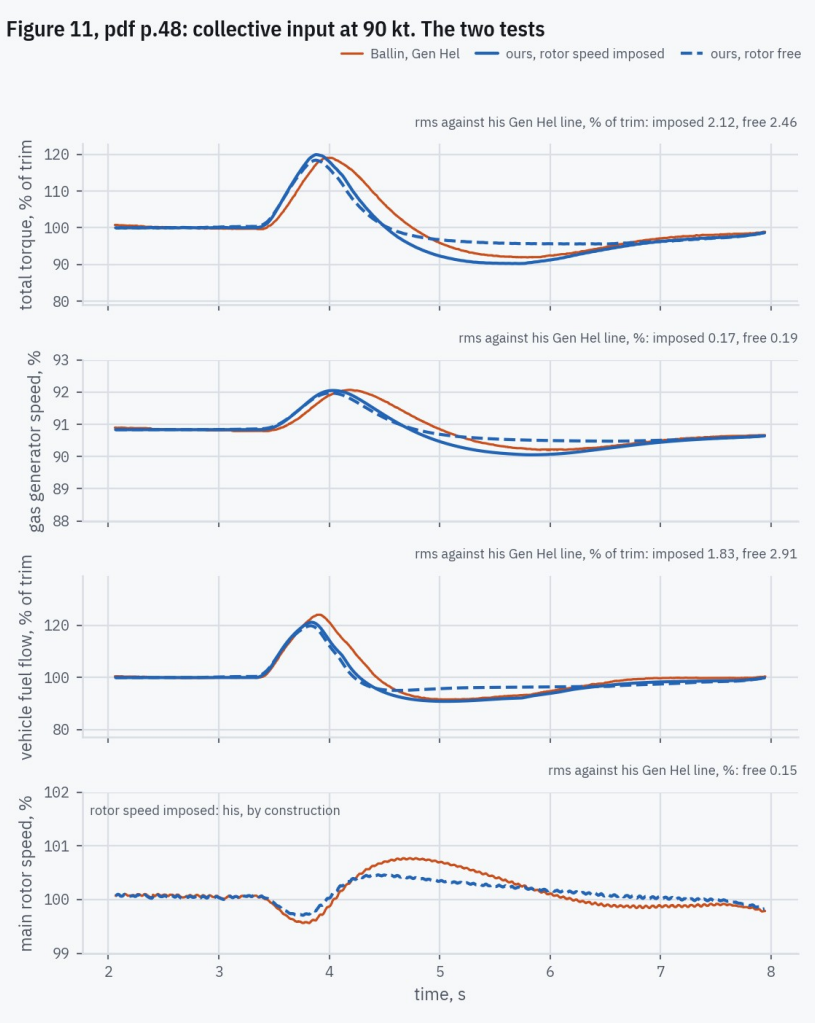


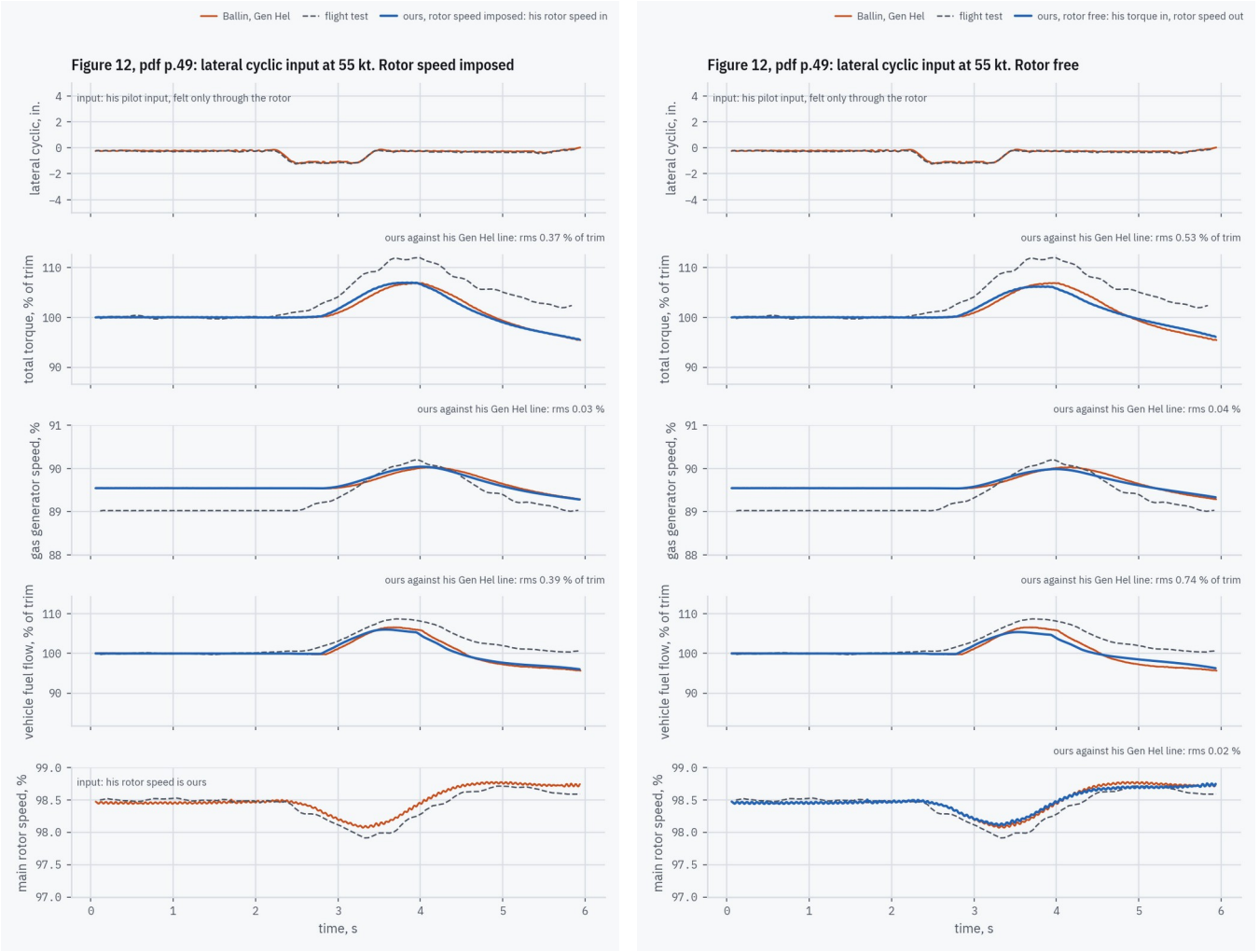
Figure 11, pdf p.48; decision log #81. Backup slide, hidden in the show

Rotor speed imposed pins his rotor speed, so a miss shows in our torque and fuel. Rotor free pins his load instead: once settled our torque must carry it, and the miss moves into our rotor speed.

Where the two methods agree, where they part, and why

- **The pulse itself:** both follow the collective up and back together.
- **After it:** they part most at 5.5 s, rotor free 5.5 % of trim above in torque. With his rotor speed imposed, our governor sees his overspeed and cuts; free, our rotor overspeeds less and the governor cuts less. His line lies between.
- **Settled** at 7.6 s: torque 97.5 and 97.4 % of trim against his 98.2.

Figure 12: lateral cyclic input at 55 kt



Left, rotor speed imposed: his rotor speed is ours; we predict torque, NG and fuel. Right, rotor free: his torque is our rotor's load; we predict the rotor speed too.

Where ours misses his, and why

- **No miss beyond a few tenths.** Rotor speed within 0.024 % rms; fuel within 0.4 % of trim (rotor speed imposed) and 0.7 % (rotor free).
- His lateral cyclic reaches the engine only through the rotor: the flight-test inputs are “single-axis” [pdf p.47], so the collective stays put.

Figure 12, pdf p.49; drive train pdf p.47; decision log #81

Figure 12: the two tests side by side



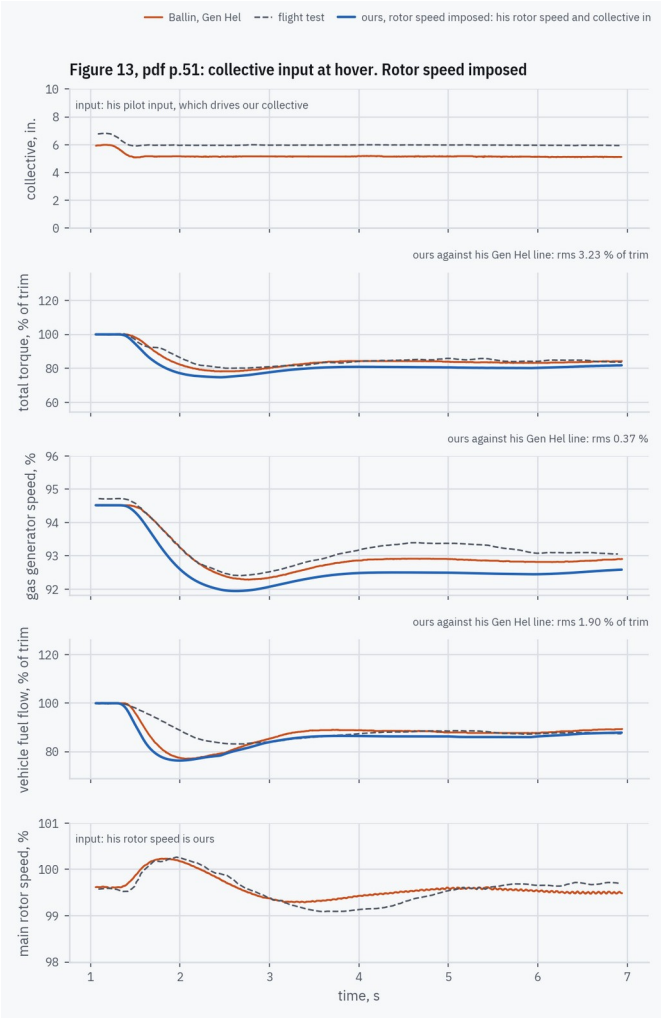
Figure 12, pdf p.49; decision log #81. Backup slide, hidden in the show

Rotor speed imposed pins his rotor speed, so a miss shows in our torque and fuel. Rotor free pins his load instead: once settled our torque must carry it, and the miss moves into our rotor speed.

Where the two methods agree, where they part, and why

- **The two agree:** 0.44 % of trim apart in torque, 0.03 % in NG, rms. A lateral cyclic input barely loads the engine, and either way our engine and fuel control answer as his do.

Figure 13: collective input at hover



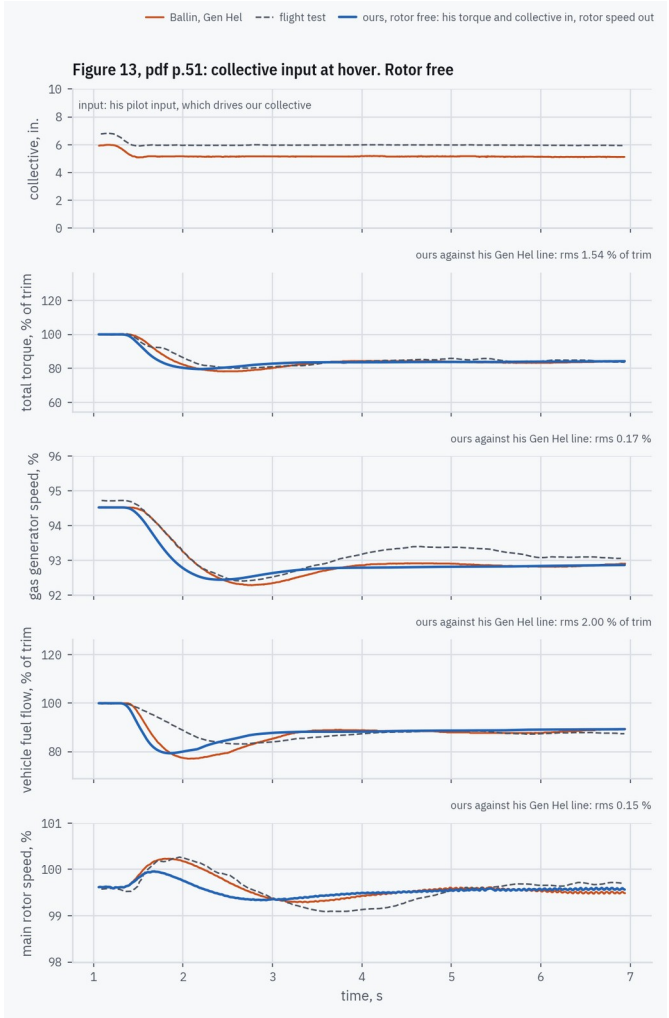


Figure 13, pdf p.51; drive train pdf p.47; decision log #81

Left, rotor speed imposed: his rotor speed is ours; we predict torque, NG and fuel. Right, rotor free: his torque is our rotor's load; we predict the rotor speed too.

Where ours misses his, and why

- **Our fuel falls sooner**, under 97 % of trim at 1.41 s against his 1.48 s, through the same load-demand path as on Figure 11.
- **Rotor speed imposed: our torque settles 3 % of trim low**: at 6 s, 80.3 % against his 83.2. Why: the collective's inch-to-percent scale is not printed. At our proportional reading, 9.6 % per inch, our collective cuts too much; at 8.0 % per inch ours reads 84.0 %. His lies between, so the scale that matches is a little below 8.8 % per inch, the bottom of the range our readings span.
- **Rotor free:** our rotor overspeeds less, 99.96 % against his 100.23 %, because our fuel falls sooner.

Figure 13: the two tests side by side

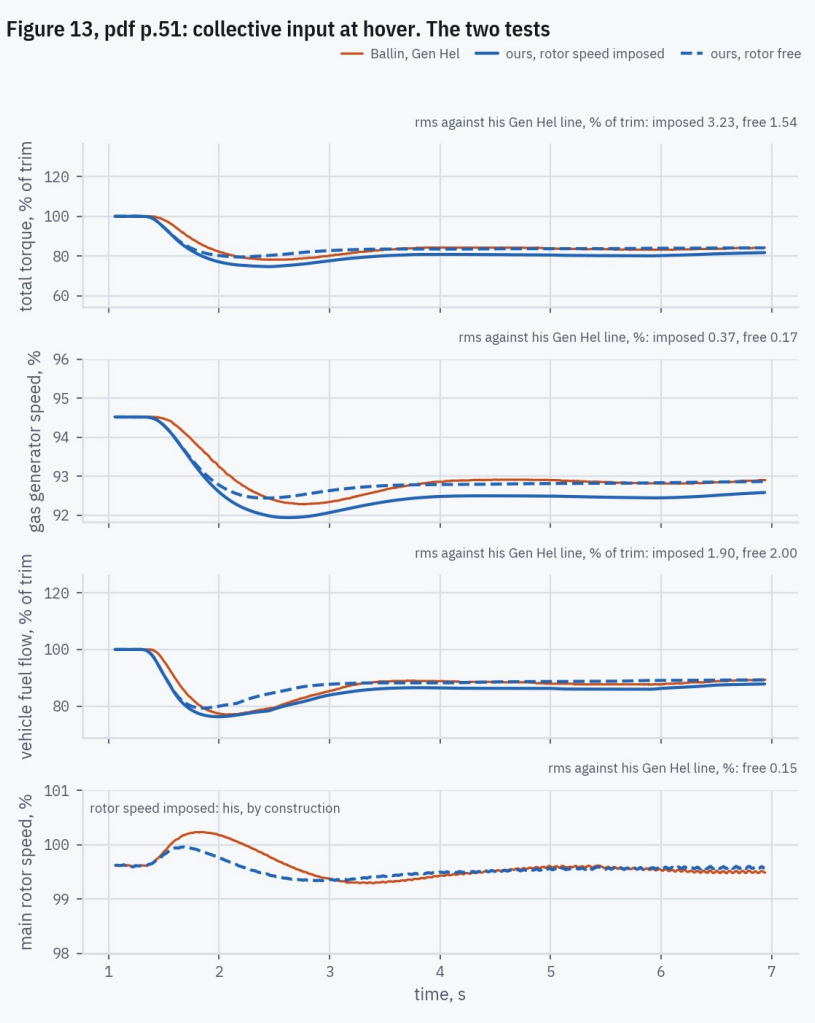


Figure 13, pdf p.51; decision log #81. Backup slide, hidden in the show

Rotor speed imposed pins his rotor speed, so a miss shows in our torque and fuel. Rotor free pins his load instead: once settled our torque must carry it, and the miss moves into our rotor speed.

Where the two methods agree, where they part, and why

- **Just after the input:** they part most at 2.7 s, rotor free 5.9 % of trim above in torque. Free, our rotor overspeeds less, because our fuel falls sooner, so the governor cuts less.
- **Settled** at 6.6 s: rotor free carries his torque, 84.1 % of trim against his 84.0, because his load sets it. With rotor speed imposed it is 81.4: there the collective's unprinted scale sets it (previous slide).

Figure 14: pedal input at hover

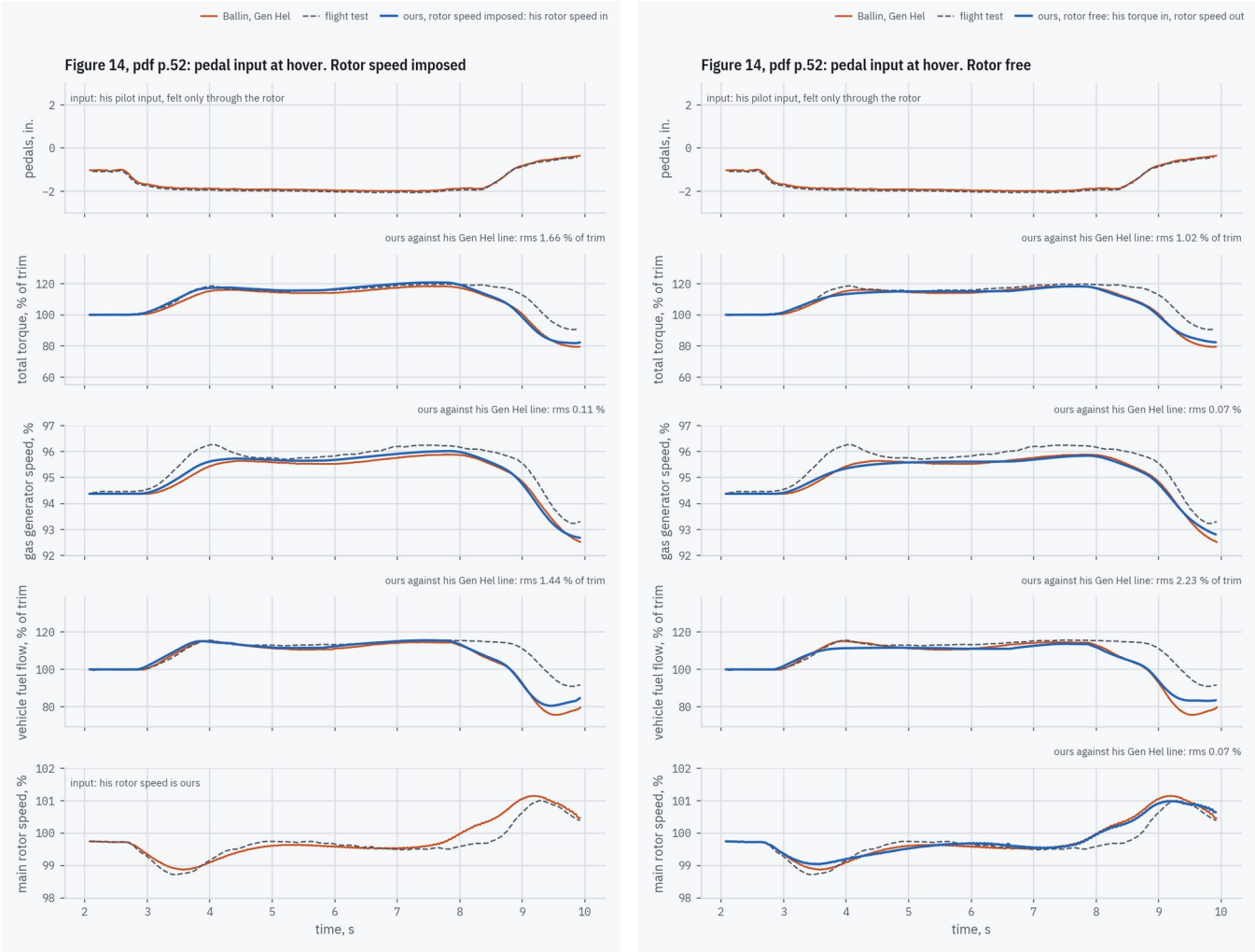


Figure 14, pdf p.52; drive train pdf p.47; decision log #81

Left, rotor speed imposed: his rotor speed is ours; we predict torque, NG and fuel. Right, rotor free: his torque is our rotor's load; we predict the rotor speed too.

Where ours misses his, and why

- **Both tests follow him closely until the end.**
- **At the end his fuel bottoms at 76 % of trim, ours at 81 %** (rotor speed imposed). As his rotor overshoots, our ECU cuts fuel, but as NG falls the HMU's droop gives part of it back. His deeper cut cannot be traced: his control's internals are not printed.

Figure 14: the two tests side by side

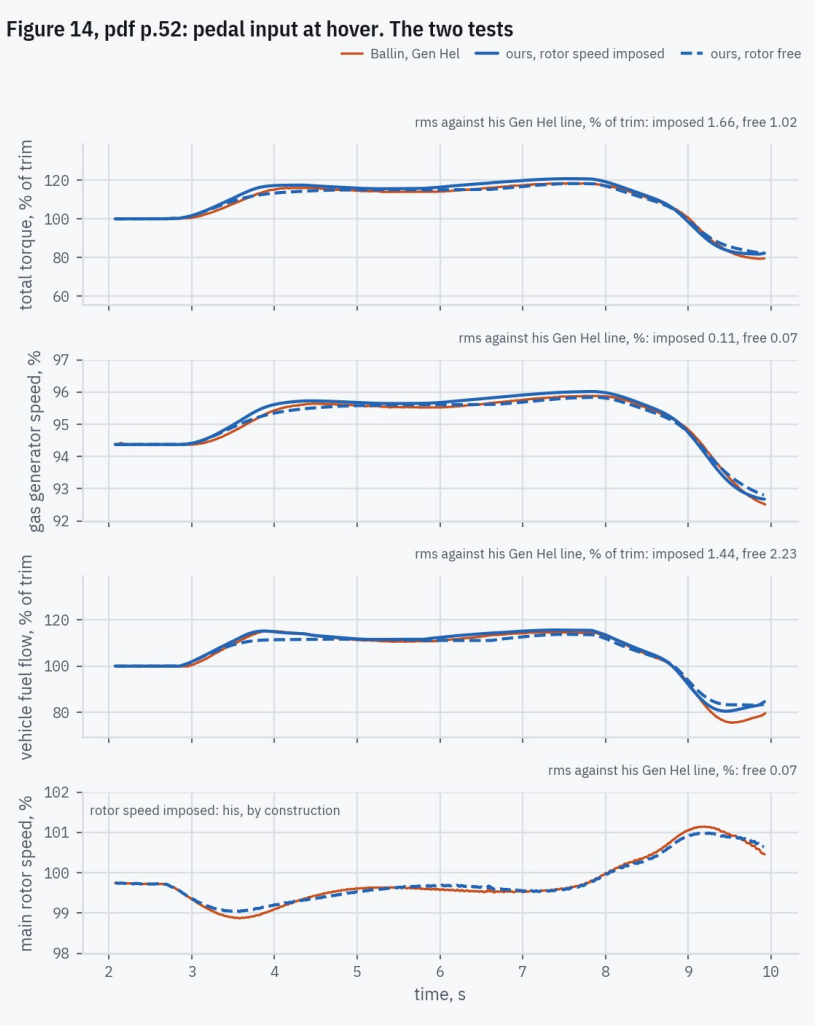


Figure 14, pdf p.52; decision log #81. Backup slide, hidden in the show

Rotor speed imposed pins his rotor speed, so a miss shows in our torque and fuel. Rotor free pins his load instead: once settled our torque must carry it, and the miss moves into our rotor speed.

Where the two methods agree, where they part, and why

- **As the pedals go in:** they part most at 3.9 s, rotor free 4.0 % of trim below in torque. Free, our rotor dips less than his, so the governor asks for less.
- **At 9.6 s**, the last reading, both sit above his fuel: 81.5 and 83.4 % of trim against his 76.1.

Figure 15: large collective input at 87 kt

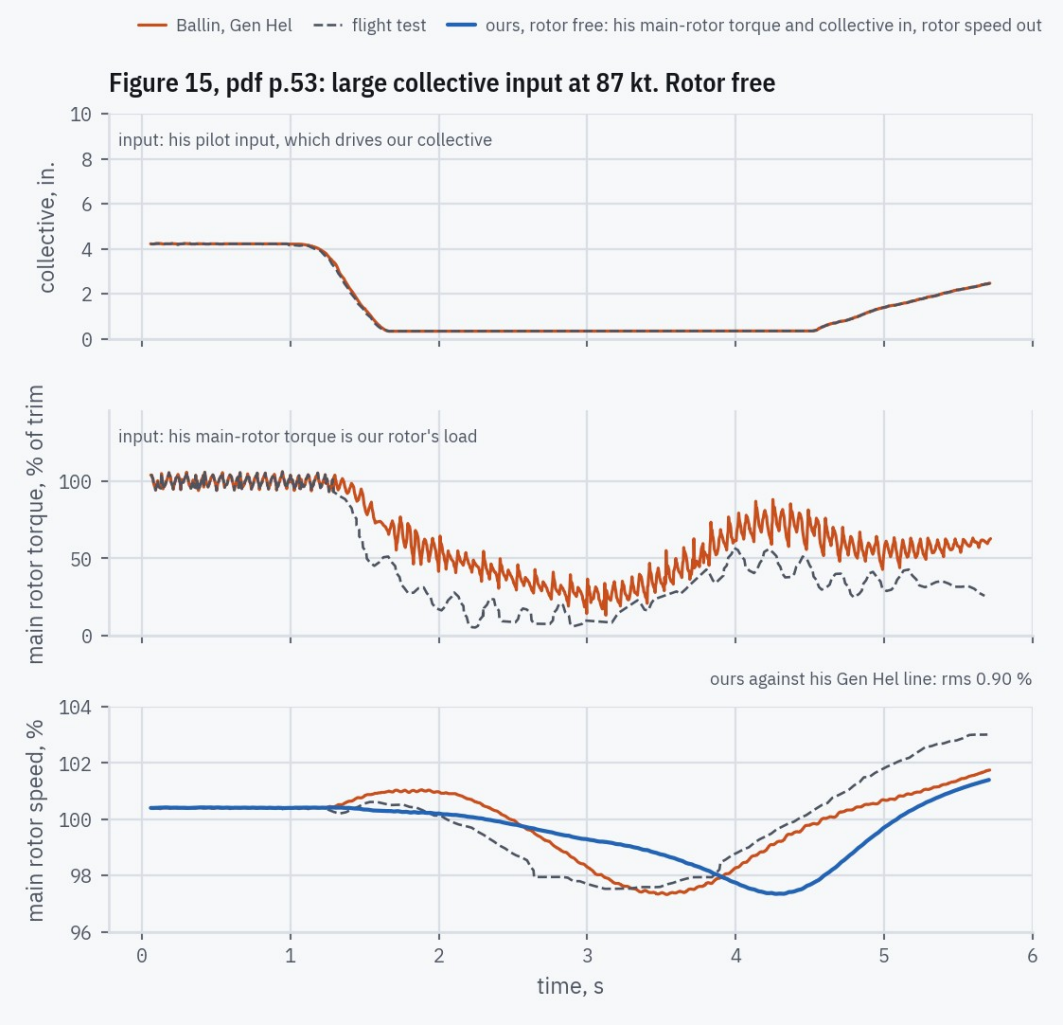


Figure 15, pdf p.53; drive train pdf p.47; decision log #81

Rotor free: his main-rotor torque is our rotor's load and his collective drives ours; our rotor makes its own speed.

Where ours misses his, and why

- **Only rotor free:** Figure 15 prints no NG, fuel or engine torque, so with his rotor speed imposed there would be nothing of his to compare.
- **Ours does not overspeed.** His rotor rises to 101.05 % at 1.89 s. Our fuel falls with the collective at once, then sits on the deceleration limit (Fig. C22, pdf p.93) from 1.5 to 2.7 s.
- **Our dip comes later.** On that limit our engine still gives torque, so our rotor slows at 1.0 %/s against his 3.2 %/s and reaches its low, 97.3 %, at 4.27 s against his 97.3 % at 3.53 s.
- **A recorded finding (decision log #81):** a flat line at the trim misses his rotor speed by 1.34 %, ours by 0.90 %: 1.49 times better, just short of the 1.5× our check asks to show that his inputs, not the trim, drive the result. Our dip is as deep as his, but 0.74 s late.

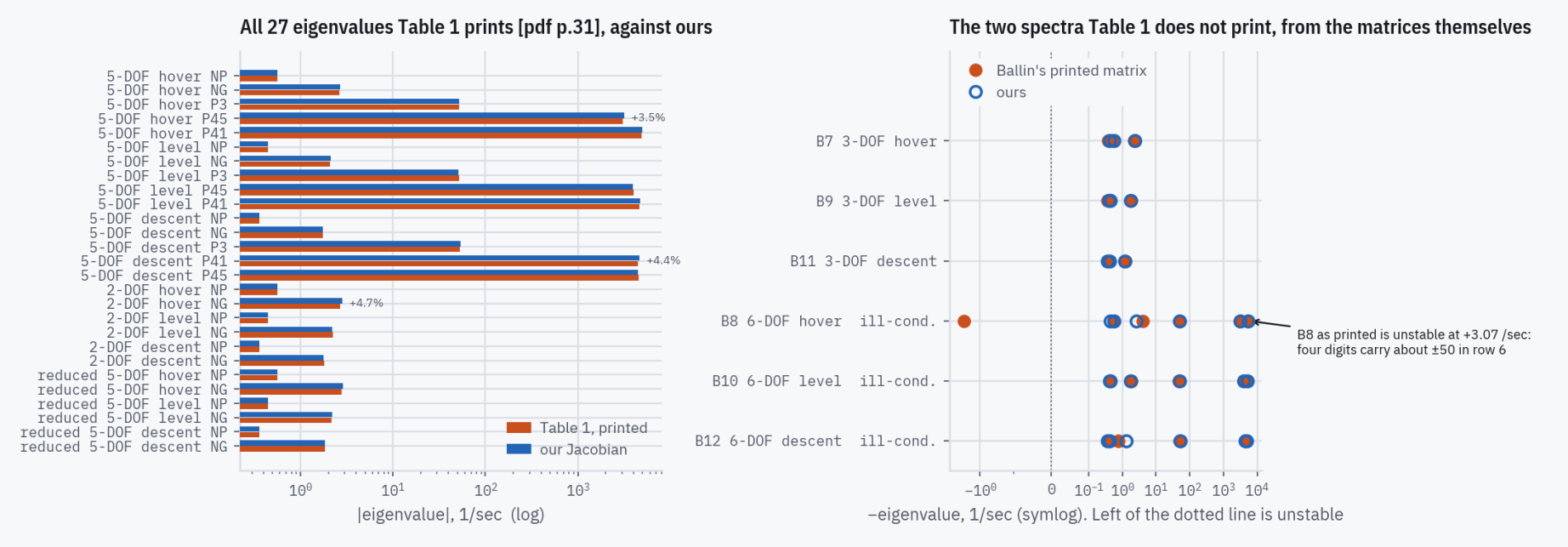
Figures 11–14: what day did they fly? Two readings, as evidence

	Standard day (used)	A: 59 °F every day	B: one common temperature
Temperature	59 °F	59 °F	79 °F
Pressure, psia (altitude, 1000 ft)	14.7 (0)	11.9–12.3 (4.8–5.8)	13.0–13.3 (2.7–3.3)
Gearbox ratio implied	—	81.7	89.0
His 8 trim numbers, rms miss	about 20 %	0.64 %	0.08 %
Rotor free, rotor speed rms, Fig 11 / 12 / 13 / 14	0.150 / 0.024 / 0.154 / 0.072	0.119 / 0.018 / 0.126 / 0.059	0.113 / 0.017 / 0.161 / 0.059

- His trims print fuel and torque at his NG. On a standard day our engine needs about a fifth more of both, on every figure: most of the gap is air pressure.
- Eight numbers, nine unknowns (a pressure and a temperature per figure, one gearbox ratio), so one assumption each: A fixes the temperature, B makes it common. B fits the trims eight times closer; A improves the transients a little more.
- **Decided:** the comparisons stay on a standard day, in % of trim. These two readings are evidence about his boundary conditions, not constants of the model.

pdf pp.47–52, 67; decision log #82

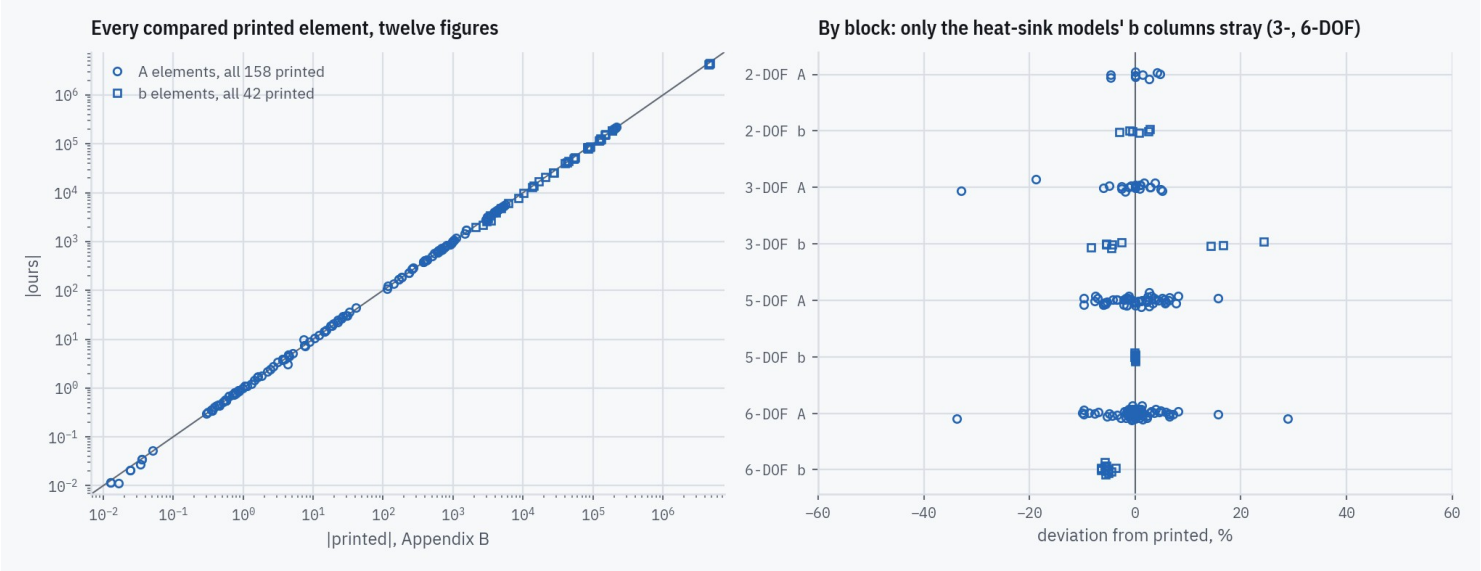
Table 1: every printed eigenvalue



- **18 tested eigenvalues** (left; NG, P3, P41, P45 modes of the 5-DOF, 2-DOF and reduced 5-DOF): rms 2.37 %, worst +4.7 % (2-DOF NG mode, hover). The 9 NP values are supplied (slide 33) and agree by construction.
- **2-DOF and reduced 5-DOF** would be one model with exact slopes; at the ±2 % step they part, Ballin's own by up to 4.3 %, ours by up to 3.5 %.
- Right: the 3-DOF and 6-DOF spectra, which Table 1 omits, against Ballin's own printed matrices.

Table 1, pdf p.31; Appendix B, pdf pp.68–76

Appendix B: every printed matrix element



Block	n	mean %	rms %	worst %
2-DOF A	6	+0.58	3.88	−4.7
2-DOF b	6	+0.24	2.02	−2.9
3-DOF A	18	−2.67	9.45	−33.0
3-DOF b	9	+2.73	11.77	+24.3
5-DOF A	51	−0.10	5.05	+15.7
5-DOF b	11	−0.02	0.08	−0.1
6-DOF A	71	−0.09	6.71	−33.7
6-DOF b	15	−5.44	5.48	−6.4

- No digitizing, no read error: the linearized model term by term against printed numbers, over eight decades.
- **187 elements tested:** 79 within 2 %, 128 within 5 %, 9 beyond 10 % (next slide). 13 are supplied, not tested: the NP diagonal and the 5-DOF hover b(NP), which set the load.
- 5-DOF fuel column: 0.08 % rms. With the heat sink (6-DOF) the same column runs uniformly low, −5.4 %. Appendix B was generated with a different heat-sink constant (Part 6): with its time constants, the NG, NP, P41 and P45 rows land within 0.2 %, and b(T41) moves to +5.5 to +6.6 %.

Appendix B, pdf pp.68–76

The largest Appendix B misses, and their causes

Of 187 tested elements, 9 miss by more than 10 %, for three reasons. None of them involves the load.

Elements	Miss	Cause	Check
3-DOF T41 row: b(T41) at all three trims, A(T41,NG) at level and descent (5)	14–33 %	Appendix B was generated with heat-sink constant $TC_{T41} \approx 0.32$; the report prints 0.29, and its transients fit 0.29 (Part 6)	With Appendix B's own time constants: b(T41) 2–6 %, A(T41,NG) 5 and 7 %
6-DOF A(P45,P3), level and descent (2)	29–34 %	A small remainder of cancelling terms: B10 and B12 print −4.315 and −7.397, where the 5-DOF prints −255.1 and −261.9	Ours collapses the same way: −3.1 and −9.9
A(NP,NG) at hover, one value printed in B2 and B8 (2)	16 %	A tiny element that changes sign across the trims (−0.02444, 0.01283, 0.03555): a small miss is a large percentage	The miss is 0.004 /s, in a row whose pressure terms are tens to hundreds

- The load enters only the NP row, and the one NP-row element here, A(NP,NG), is an engine torque derivative: the load does not depend on NG.
- Supplied, not tested: the 13 elements that set the load (slide 33).

Appendix B, pdf pp.68–76; Table A.1 (TC_T41), pdf p.55; Eqs. 51–53, pdf p.26

Why slopes agree less well than levels

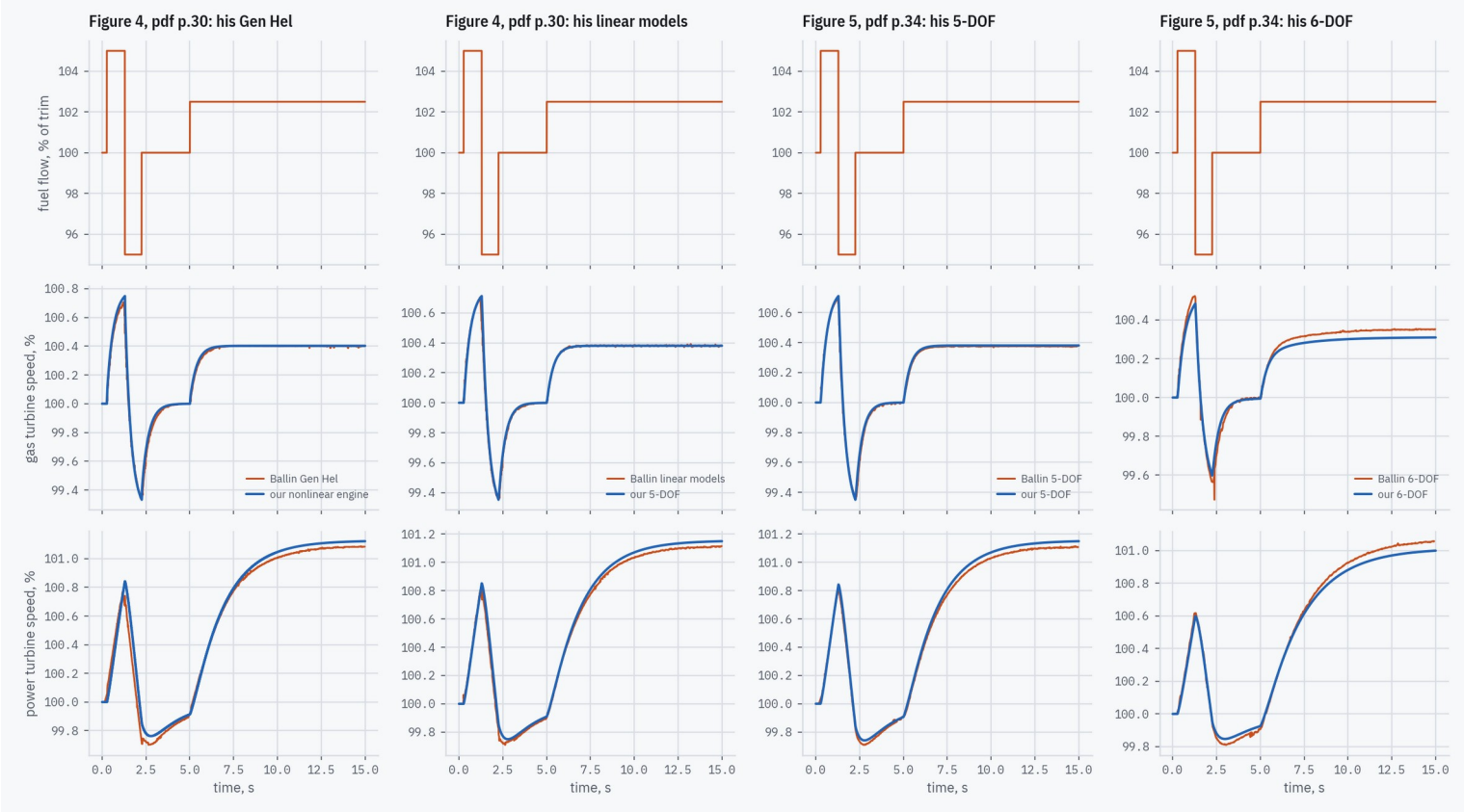
- **Levels:** the trims match Table B.1 to 0.22 % rms. **Slopes:** Table 1 to 2.37 % rms, Appendix B's elements to 6.5 %.
- A linear model is a table of map slopes. Every map is read to about a tenth of a percent of its span, but a slope differences two readings close together, and it jumps at each breakpoint of a piecewise-linear table.
- **The step matters most.** Ballin's ±2 % is an amplitude: it averages across the tables' kinks. Using it as he did is what brings Table 1 to 2.4 %.
- **Which compressor table is used barely matters** for Table 1: our traced lines and Ballin's seven-point table agree to a tenth of a point.

How the slopes are taken	rms	worst
Ballin's ±2 % step, our traced compressor lines (as built)	2.37 %	4.7 %
The same, with Ballin's seven-point compressor table	2.34 %	4.4 %
A tiny step (10^{-5}) instead of ±2 %	6.62 %	14.9 %
Smooth curves instead of straight lines between the same printed points*	5.85 %	16.1 %

The 18 tested Table 1 eigenvalues. *Decision log #43, measured 2026-10-12 over all 27 modes, when straight lines gave 1.75 %.

Method pdf p.28; decision log #43, #56, #60

Figures 4–5: Ballin's linear-model curves against ours



Ballin's curve	Ours	NG / NP
Fig 5: 5-DOF	5-DOF	0.005 / 0.028
Fig 5: 6-DOF	6-DOF	0.030 / 0.033
Fig 4: Gen Hel	nonlinear	0.008 / 0.045
Fig 4: linear	5-DOF	0.005 / 0.025
Fig 4: linear	2-DOF	0.012 / 0.011
Fig 4: linear	reduced	0.005 / 0.026

rms difference, % of trim

- **Figure 5:** our 5-DOF lies on his 5-DOF; our 6-DOF follows his 6-DOF, slightly more damped.
- **Figure 4:** his three linear models lie on one another; his Gen Hel line parts from them. Our linear models follow his linear line.
- In NP his Gen Hel line leads and dips at each fuel step: rotor-blade lag, which the linear rotor load leaves out (pdf p.29). Our nonlinear engine carries the same linear rotor load, so in NP it follows his linear line (0.012 %).
- His own printed hover matrices draw his curves (0.003 % on Figure 5's 5-DOF): the trim is hover.

Figures 4–5, pdf pp.30, 34; text pp.29, 33; Appendix B, pdf pp.68–70

Values the report never prints, recovered from its own numbers

- **Heat-sink time constants** τ_1, τ_2 (Eq. 63): printed nowhere. Each 3-DOF block of Appendix B fixes both: 2.080 / 3.099 s (hover), 2.200 / 3.413 s (level), 2.390 / 3.854 s (descent).
  - They need $TC_{T41} \approx 0.32$, not the printed 0.29: **Appendix B was generated with a different constant than the transient simulations.** We keep the printed 0.29; the report's transients fit it better.
- **Load inertia and load slope** (Eqs. 46–47): owned by the helicopter model. Recovered from Appendix B's NP rows: $J_{load} = 9.25 \times J_{PT}$.
- **Standard day** (Eqs. 5, 8): T_{std}, P_{std} never printed; 518.67 °R and 14.696 psia, consistent with Table B.1.
- **Power-turbine speed for each comparison:** design speed, 20,900 rpm, for Figures 6–10; 20,895 rpm for Table B.1, Table 1 and Appendix B, as Table B.1 prints.
- **The fuel-step instants** of Figures 9–10, read off the figures' own fuel panels.

Eq. 63, pdf p.31; Appendix B, pdf pp.68–76; Table A.1, pdf p.55

Problems in the printed method, found and handled

- **Eq. 80's P45 iteration cannot converge in a fuel chop.** Its fixed point repels where f9's slope (in log terms) is steeper than −1. Below flight idle the frame then sat on false equilibria up to 7 %NG away from the true ones. We solve the same equation exactly, which takes one division.
- **The heat sink is what fixed the transient.** Without it our T41 spike overshot 3.3 times as far as Ballin's. With the heat sink integrated as the metal temperature (Eqs. 48–49) it sits at 0.6–1.2 times his.
- **“±2 %” is an amplitude, not a derivative step.** Using it as Ballin did took Table 1 from 6.6 % to 2.4 % rms.
- **Station 4.5 mixing does not conserve energy**: Eq. 29 makes the mixed enthalpy a fixed fraction of H44 [pdf p.24]. The printed coefficient sits 1.1–1.7 % below an energy-conserving mix. Reproduced as printed.
- **The closed-loop limit cycle** is the printed NG-sensor hysteresis, not a numerical artifact.

Eq. 80, pdf p.37; Eqs. 48–53, pdf pp.25–26; pdf p.28; Eq. 29, pdf p.24

Misprints, and how the report settles them

Where	As printed	What the rest of the report requires
Eq. 43	fuel leaves the 4.1 volume ($-W_f$)	$+W_f$, as Eq. 73 prints the same balance
Eqs. 79–80	$B_3 \cdot B_4 \cdot WA2$; the engine model has no B_4	$B_3 \cdot K_{bl} \cdot WA2$, as Eq. 44 writes it
Eq. 74	lags only the bleed flow	the sentence above it lags the whole compressor flow; both coded, prose used
pdf p.32	F2's sixth column labelled as another state	the column is with respect to T41
Table A.1	units of TC_{T41}	Eq. 51 is right; the printed units are not
Fig. A10	axis labelled PS9/P49	the values are P49/Ps9, as Eq. 38 needs
Fig. A3	the 0.11 tick label sits off its tick	the plateau is 0.1091; the label is the defect
Table 1	descent modes P41 −4430, P45 −4530	Figure B6, the same model and trim, has them the other way round; scored as printed

In each case the report itself decides the question; nothing is corrected from outside knowledge.

pdf pp.25–26, 31–32, 36–37, 55, 58, 65, 70

Where we differ from the report: four departures

What	The report	We	What it changes
P45 solve (Eq. 80)	fixed-point iteration	the same iteration first; an exact solve where it cannot converge	only in a chop and below flight idle: removes false equilibria
Compressor map interpolation (f1)	seven printed points per speed line	each speed line traced from the drawn line; his table kept beside it	better on some comparisons, worse on others; kept by decision
Above the map's top line	the map ends at 100 %NG	f1 extends a tenth of a point above 100 % before it clamps	reaches Ballin's top points of Figures 6–8; nothing else moves
Limited integrators (Figs. C3, C7, C10)	an integrator and a separate limiter, which winds up	a clamped integrator, which does not	identical on everything printed; differs only in a sustained load dump

pdf pp.37, 39, 56, 86–89

Where the report is silent or ambiguous, and what we chose

Question	Our choice	Why
Which runs used the heat sink	a switch; Figures 9–10 with it	the report says so only for its linear results; for 9–10 it is our inference
Order inside a frame (Q_{GT}'s P45)	the previous frame's P45	Ballin's own Figure 9 fits it clearly better
Eq. 74's two readings	the prose: whole flow lagged	it is the sentence that describes opening the loop; both converge to the same trim
Combustor efficiency f6	a constant	Figure A6 draws one; the slope was the page's tilt
Hysteresis constants	the whole band	the ordinary meaning; no printed response isolates it
Integrator; table interpolation	explicit Euler; linear	neither is named; linear is what the report's own numbers show
Initial state of the fuel control	every block in equilibrium	a trimmed engine then sits still

Limits of the source data, and how questions were tracked

- **Two questions stay open on purpose.** Neither blocks a result; both concern the compressor map, which sits under every comparison:
  - **How to interpolate the map.** Ballin's seven-point table and our traced lines each win on some comparisons; the choice is recorded as a judgement.
  - **Figure A1's eleven speed lines do not form one smooth family** within our measured read error. Either the read error is understated across lines, or the printed lines are not one smooth family. Nothing in the report can settle it.

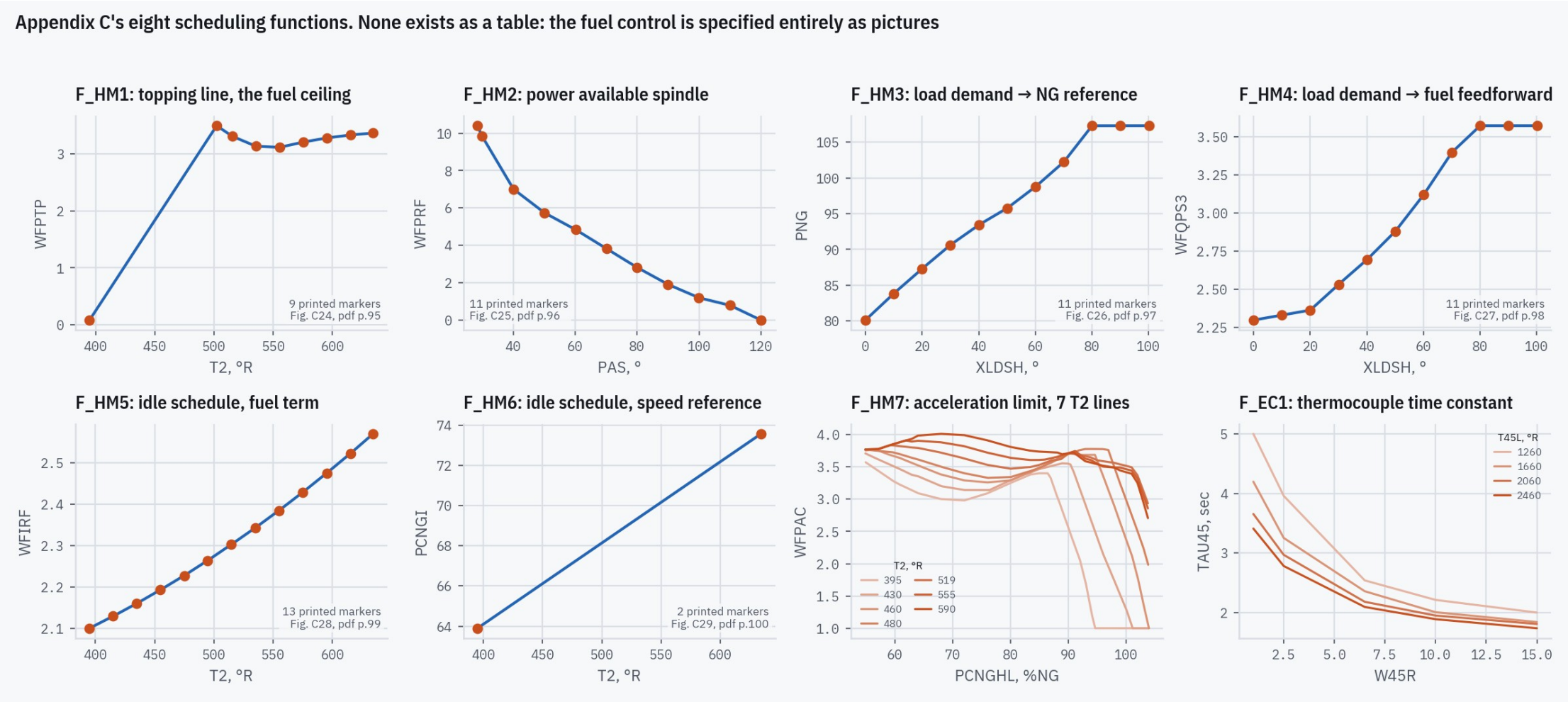
The decision log: every question the report raised was written down with its evidence and its outcome.
79 logged · 77 settled · 2 kept open on purpose

Backup slide, hidden in the show

Summary

- A real-time engine model is a **deliberate approximation** of the nonlinear one: the slow dynamics (spools, heat soakage, controls, sensors) are integrated; the fast gas-volume dynamics are solved inside each 7 ms frame.
- The replication reproduces every comparison the report's printed data allows: trims to 0.22 % rms, eigenvalues to 2.4 % over the 18 tested modes, steady sweeps and transients close to Ballin's own lines.
- With the fuel control closed, the engine finds Table B.1's printed state on its own (worst 0.25 %) and follows Ballin's Figures 11–15 manoeuvres; where it parts from his line, the cause is measured in our control.
- Values the report never prints were recovered **from its own numbers**, never from outside knowledge, and some of them show where the report disagrees with itself.
- Four departures, each measured; everything else follows the report, including its known biases.

Appendix C's eight schedules, as digitized



Figures C23–C30, pdf pp.94–101

Table B.1 in full

Trim	Quantity	Printed	Ours	Dev. %
hover	NG	41,638	41,638	+0.000
hover	PS3	176.34	176.18	−0.089
hover	P41	174.28	174.14	−0.081
hover	T41	2,292	2,292	+0.007
hover	P45	37.42	37.40	−0.061
hover	T45	1,632	1,633	+0.069
hover	SHP	911.1	910.6	−0.060
level 80 kt	NG	39,768	39,816	+0.120
level 80 kt	PS3	142.13	142.11	−0.013
level 80 kt	P41	140.26	140.24	−0.011
level 80 kt	T41	2,102	2,101	−0.024
level 80 kt	P45	30.66	30.65	−0.039
level 80 kt	T45	1,501	1,500	−0.049
level 80 kt	SHP	552.6	552.2	−0.077
descent 80 kt	NG	38,072	38,054	−0.048
descent 80 kt	PS3	114.27	113.88	−0.338
descent 80 kt	P41	112.77	112.40	−0.331
descent 80 kt	T41	1,982	1,984	+0.087
descent 80 kt	P45	25.54	25.47	−0.265
descent 80 kt	T45	1,424	1,426	+0.172
descent 80 kt	SHP	302.6	300.2	−0.806

Table B.1, pdf p.67

Table 1 in full

Model	Trim	Mode	Printed	Ours	Dev. %
5-DOF	hover	NP	−0.565	−0.565	supplied
5-DOF	hover	NG	−2.66	−2.69	+1.18
5-DOF	hover	P3	−51.6	−51.7	+0.19
5-DOF	hover	P45	−3,060	−3,166	+3.47
5-DOF	hover	P41	−4,900	−5,007	+2.18
5-DOF	level	NP	−0.446	−0.446	supplied
5-DOF	level	NG	−2.08	−2.11	+1.51
5-DOF	level	P3	−52.2	−51.0	−2.35
5-DOF	level	P45	−4,040	−3,948	−2.28
5-DOF	level	P41	−4,640	−4,736	+2.08
5-DOF	descent	NP	−0.357	−0.357	supplied
5-DOF	descent	NG	−1.75	−1.75	+0.28
5-DOF	descent	P3	−52.6	−53.6	+1.83
5-DOF	descent	P41	−4,430	−4,624	+4.38
5-DOF	descent	P45	−4,530	−4,445	−1.87
2-DOF	hover	NP	−0.565	−0.565	supplied
2-DOF	hover	NG	−2.69	−2.82	+4.70
2-DOF	level	NP	−0.446	−0.446	supplied
2-DOF	level	NG	−2.23	−2.20	−1.27
2-DOF	descent	NP	−0.357	−0.357	supplied
2-DOF	descent	NG	−1.82	−1.77	−2.85
reduced 5-DOF	hover	NP	−0.565	−0.565	supplied
reduced 5-DOF	hover	NG	−2.81	−2.86	+1.82
reduced 5-DOF	level	NP	−0.446	−0.446	supplied
reduced 5-DOF	level	NG	−2.16	−2.21	+2.11
reduced 5-DOF	descent	NP	−0.357	−0.357	supplied
reduced 5-DOF	descent	NG	−1.83	−1.83	+0.14

Table 1, pdf p.31

Glossary

Symbol	Meaning	Symbol	Meaning
NG, NP	gas-generator, power-turbine speed	Wf, WF	fuel flow
%NG	NG as % of design, 44,700 rpm	WA2, WA3	air flow at stations 2, 3
NGc	corrected gas-generator speed	W41, W45	gas flow into the turbines
P3, P41, P45	total pressures at stations 3, 4.1, 4.5	Ps3	compressor static discharge pressure
T41, T45	temperatures at 4.1, 4.5 (T45: PT inlet)	θ_2, δ_2	T2, P2 over the standard day
Q_{GT}, Q_C, Q_{PT}	turbine, compressor, PT torques	Q_{req}	load torque (helicopter)
J_{GT}, J_{PT}, J_{load}	inertias	K_{V3}, K_{V41}, K_{V45}	volume coefficients
ECU, HMU	electrical, hydromechanical control units	SPDG	ECU output to the torque motor
DOF	degrees of freedom (states) of a linear model	SHP	shaft horsepower
H2 … H49	enthalpies at the stations, Btu/lbm	θ_{41}, θ_{45}	squared critical velocity ratios (Eqs. 25, 31)

Nomenclature, pdf pp.9–13; Appendix C nomenclature, pdf pp.77–82